BT301 FINAL TERM

PPT Slide 96 TO 209

Merged File date 2020

Regards : **Zarva Chaudhary**

Chaudhary Moazzam

Plasmid Biology

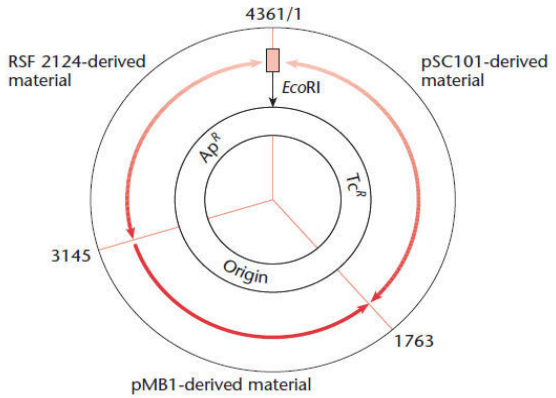
Use of pSC101 for cloning

- Expression of *Staphylococcus* plasmid genes in *E. coli*
- pSC101 contain Tetracycline resistance *P1248 from Staphylococcus*

Plasmid Biology

pBR322, a purpose-built cloning vehicle

- pBR322 is an example of *in vitro* constructed cloning vehicle
- pBR322 contains the Ap^R and Tc^R genes of RSF2124 and pSC101, combined with replication element of pMB1



The origin of plasmid pBR322

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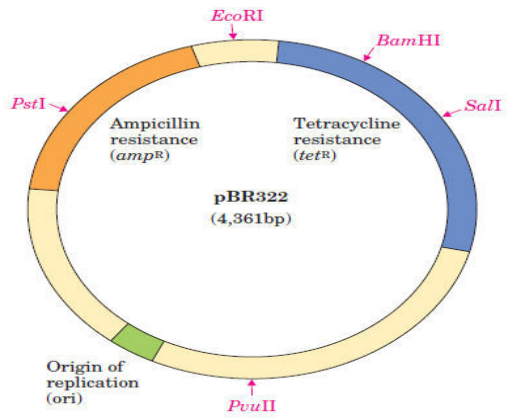
Plasmid Biology

pBR322, a purpose-built cloning vehicle

- It has 40 target sites for different restriction enzymes

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The constructed E. coli plasmid pBR322

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Plasmid Biology

pBR322, a purpose-built cloning vehicle

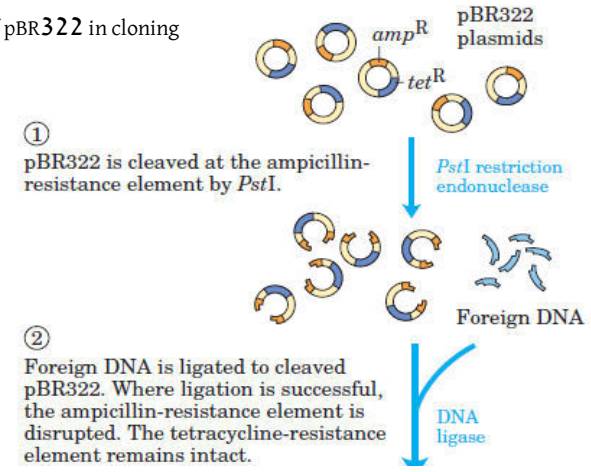
- It is difficult to clone DNA segments longer than about 15,000 bp when plasmids are used as the vector

Plasmid Biology

Cloning with pBR322

- Plasmid pBR322 has a number of characteristics that make it suitable as cloning vehicle

Use of pBR322 in cloning

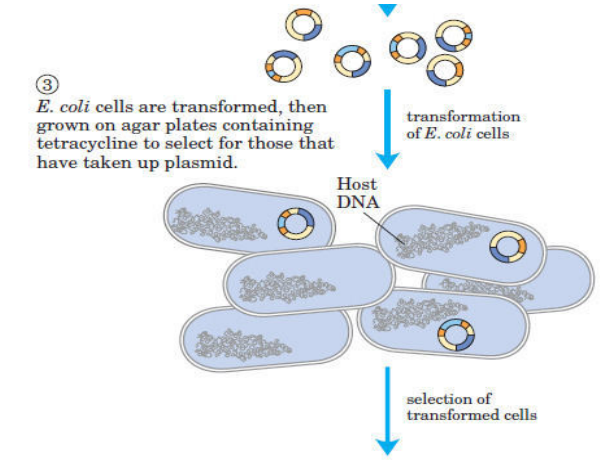


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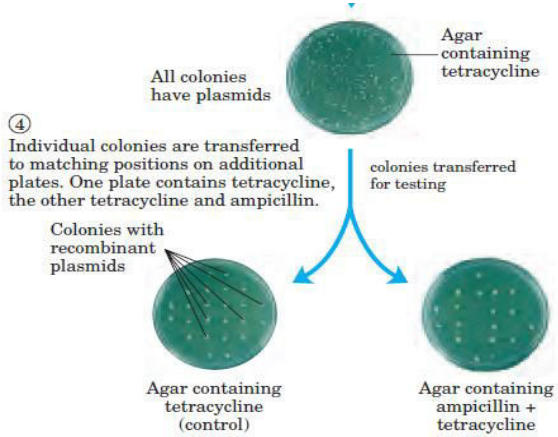
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Use of pBR322 in cloning



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Use of pBR322 in cloning



Plasmid Biology

Improved vectors derived from pBR322

- Over the years numerous different derivatives of pBR322 have been constructed
- pBR325 encodes chloramphenicol resistance in addition to ampicillin and tetracycline

Plasmid Biology

Improved vectors derived from pBR322

- pUC19 carrying sites for many different restriction endonucleases
- pAT153 another derivative of pBR322

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Plasmid Biology

Runaway plasmid vectors

- Loss of copy number control -runaway replication

Plasmid Biology

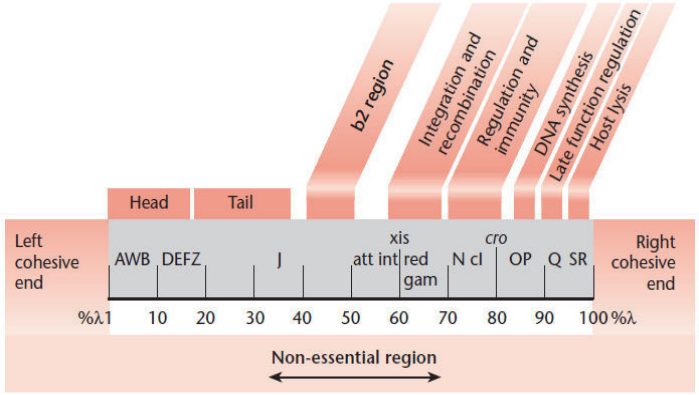
Runaway plasmid vectors

- Loss of copy number control -runaway replication

Phage vectors

Bacteriophage ʎ as cloning vector

- Bacteriophage ʎ has a very efficient mechanism for delivering its 48,502 bp of DNA into bacterium
- About one-third of the ʎ genome is nonessential and can be replaced with foreign DNA



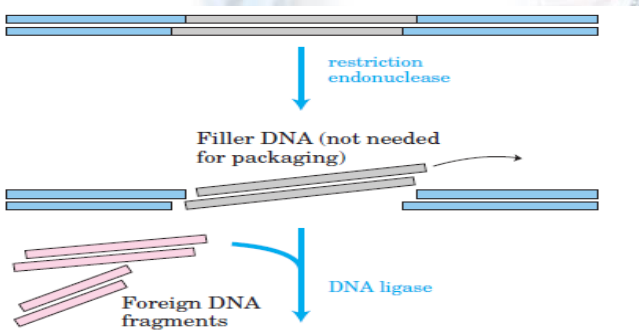
Map of ʎ chromosome

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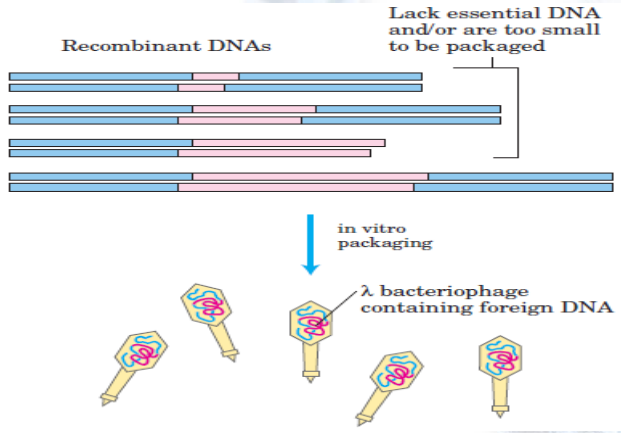
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Phage vectors



Bacteriophage ʌ cloning vectors

Phage vectors



Bacteriophage ʌ cloning vectors

Phage vectors

Bacteriophage ʌ as cloning vector

- Bacteriophage ʌ permits the cloning of DNA fragments of up to 23,000 bp

Phage vectors

Replication of phage-ʌ DNA in lytic and lysogenic cycles

- The lytic cycle results in the destruction of the infected bacterial cells
- Viral DNA exists as separate molecule within the bacterial cell and replicates independently of the host DNA

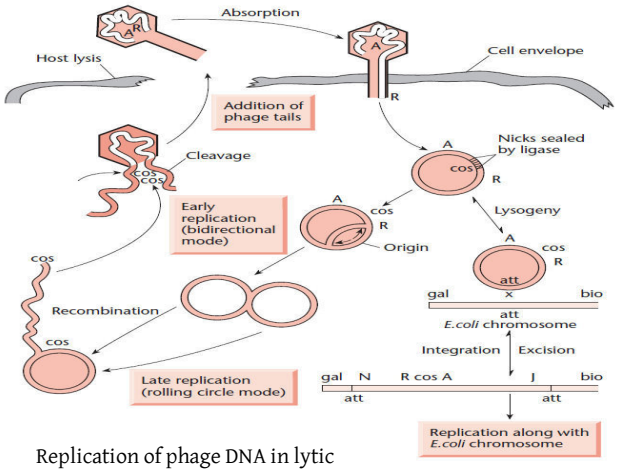
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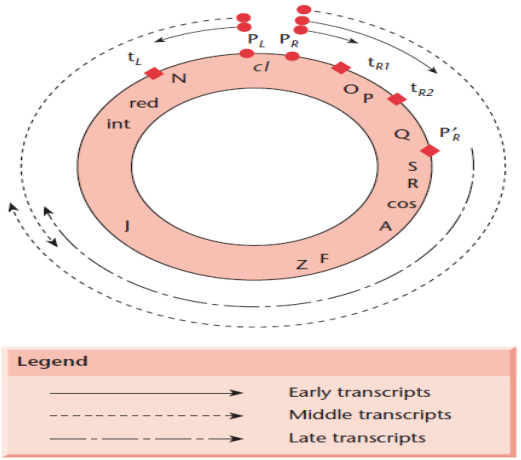
Phage vectors

Replication of ʎ DNA in lytic and lysogenic cycles

- Lysogeny is characterized by the integration of the ʎ DNA into the host bacterium genome



Replication of phage DNA in lytic and lysogenic cycles



Major promoters and transcription termination sites of ʎ

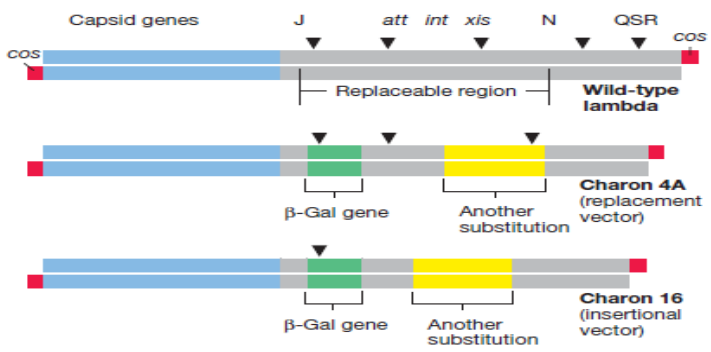
Phage vectors

Modified ʎ phages

- Wild type is not suitable as a cloning vector because it has too many restriction enzymes sites
- Insertional vectors
- Replacement vectors

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Modified ʎ cloning vectors

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Phage vectors

Modified ʎ phages

- Charon 4A and 16 are both derivatives of lambda with various substitutions and deletions in the nonessential region
- Each has the *lacZ* gene, encoding enzyme β-galactosidase, which permits detection of phage containing inserted clones

Phage vectors

Steps in cloning with ʎ

- Cloning with lambda replacement vectors involves the following steps

(i). Isolation and cutting vector DNA with appropriate restriction enzymes

(ii). Connecting the two lambda fragments to foreign DNA by using DNA ligase

Phage vectors

Steps in cloning with ʎ

(iii). *In vitro* packaging of recombinant DNA

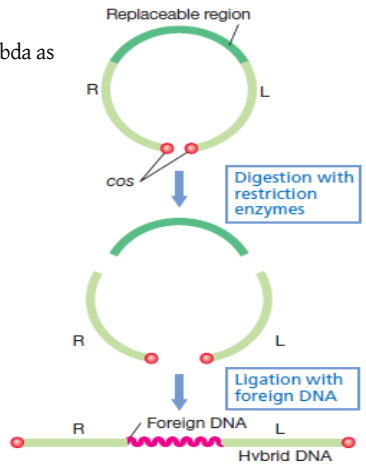
(iv). Infection of *E. coli* cells and isolation of phage clones by picking plaques on a host strain

(v). Checking recombinant phage for the presence of foreign DNA

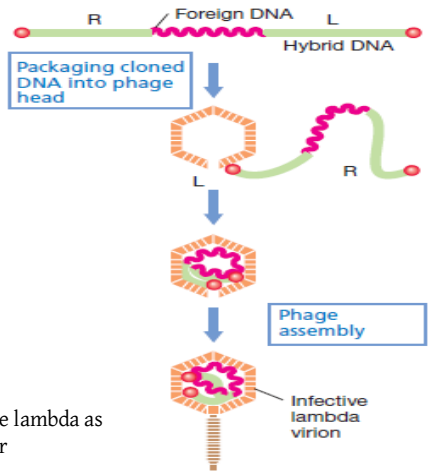
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Bacteriophage lambda as cloning vector



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Bacteriophage lambda as cloning vector

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Phage vectors

Steps in cloning with ʎ

- Both Charon vectors are also engineered to contain reporter genes such as for β-galactosidase for detection of recombinant phages

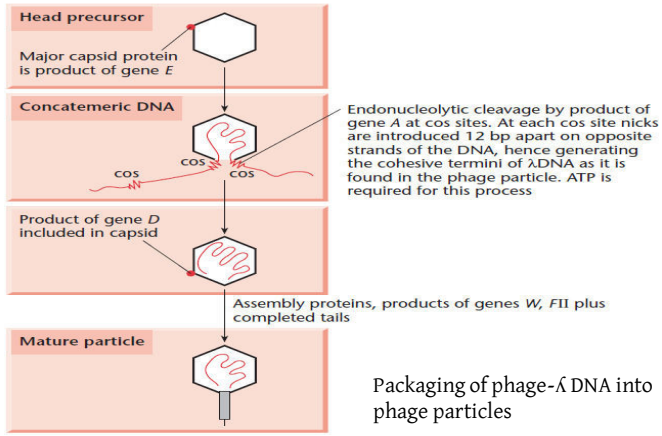
Phage vectors

Packaging phage-λ DNA *in vitro*

- Recombinant DNA in a phage coat allows it to be introduced into the host bacteria by the normal processes of phage infection
- Packaging *in vitro* yields about 10^6 plaques/ µg of vector DNA

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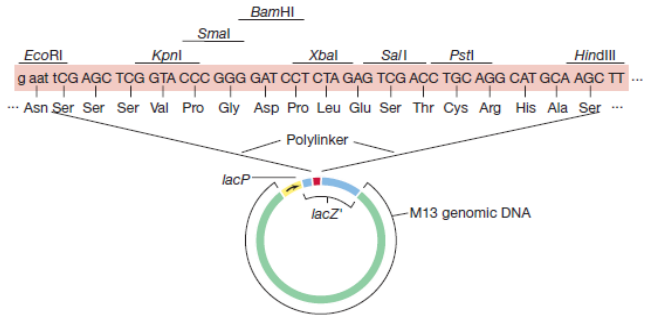
Packaging of phage-ʎ DNA into phage particles

Phage vectors

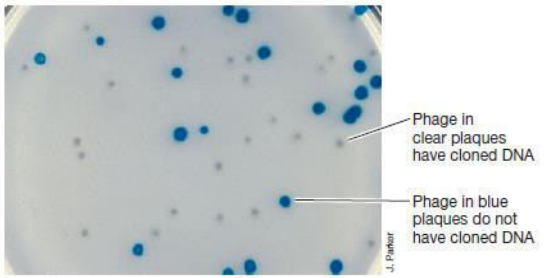
Vectors for DNA sequencing:

bacteriophage M13

- M13 is a filamentous bacteriophage with single stranded DNA that replicates without killing its host
- Variable lengths of DNA upto about 5 kbp, can be cloned



Cloning using bacteriophage M13mp18



Cloning using bacteriophage M13mp18

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Phage vectors

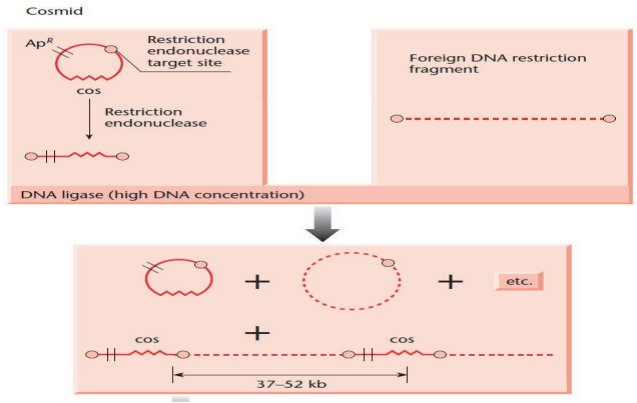
Vectors for DNA sequencing: bacteriophage M13

- Use of M13 in molecular cloning

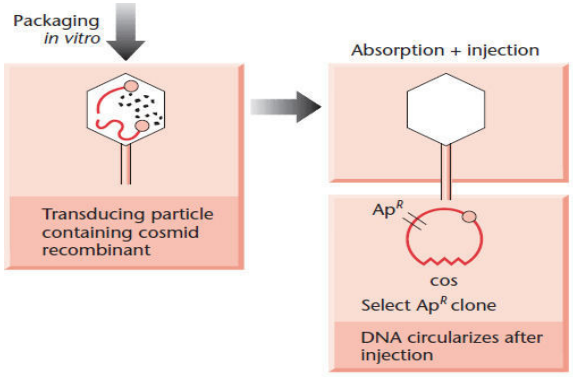
Cosmids, phasmids and other vectors

Cosmid vectors

- Plasmids have been constructed which contain a fragment of λ DNA including '*cos*' site
- These plasmids have been termed cosmids and can be used as gene-cloning vectors in conjunction with the *in vitro* packaging system



Simple scheme for cloning in a cosmid vector



Simple scheme for cloning in a cosmid vector

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Cosmids, phasmids and other vectors

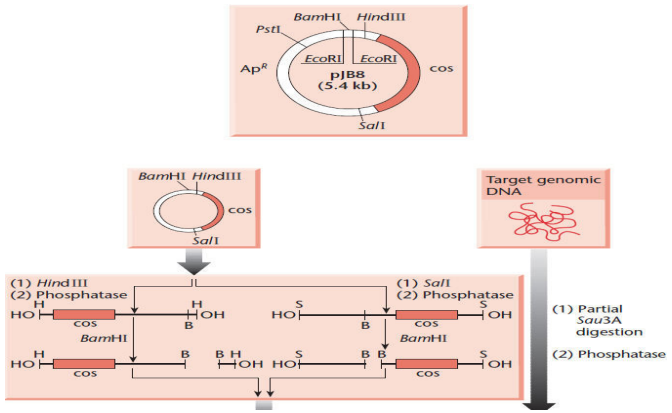
Cosmid vectors

- Cosmids provide an efficient means of cloning large pieces of foreign DNA
- Cosmids are particularly attractive vectors for constructing libraries of eukaryotic genomic fragments

Cosmids, phasmids and other vectors

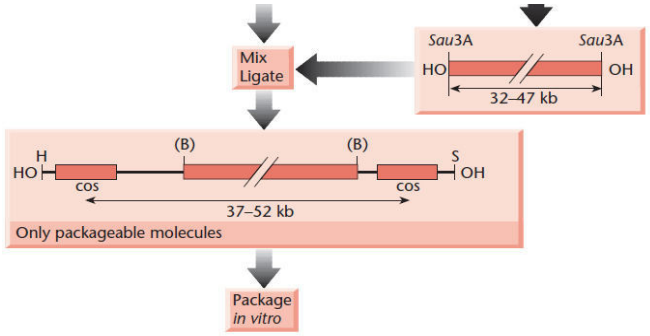
Modified schemes for cloning in Cosmid vectors

- Difficulties in cloning with cosmids have been overcome in a cosmid cloning procedure devised by Ish-Horowicz and Burke (1981)



Plasmid cloning scheme of Ish-Horowicz and Burke (1981)

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Plasmid cloning scheme of Ish-Horowicz and Burke (1981)

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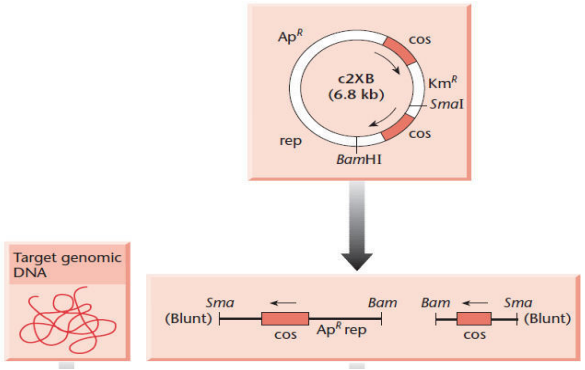
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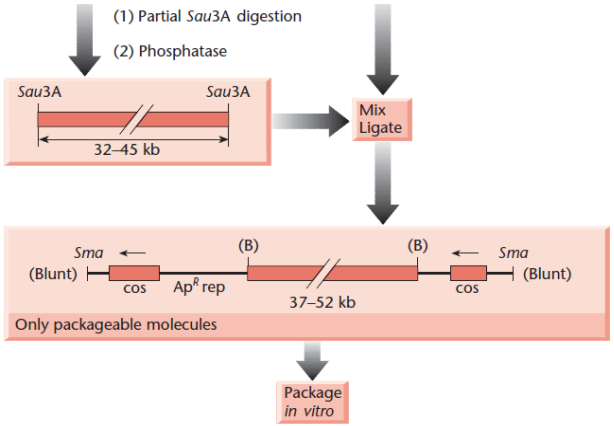
Cosmids, phasmids and other vectors

Modified schemes for cloning in Cosmid vectors

- An alternate solution to these problems has been devised by Bates and Swift (1983) Who constructed cosmid c2XB



Cosmid cloning scheme of Bates and Swift (1983)



Cosmid cloning scheme of Bates and Swift (1983)

Cosmids, phasmids and other vectors

Phasmid vectors

- Phasmid vectors are combination of plasmid and λ phage sequences
- It consists of a plasmid vector carrying a λ attachment (λatt) site

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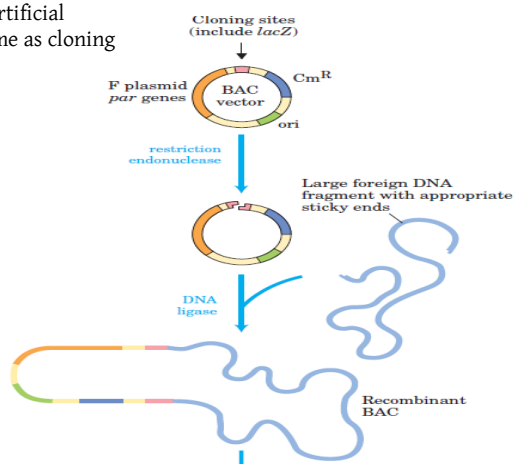
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Cosmids, phasmids and other vectors

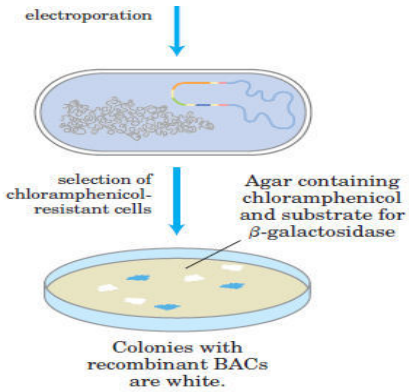
Bacterial artificial chromosomes (BACs)

- Bacterial artificial chromosomes are simply plasmids designed for the cloning of very long segments of DNA (100,000 to 300,000 bp)

Bacterial artificial chromosome as cloning vectors



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Bacterial artificial chromosome as cloning vectors

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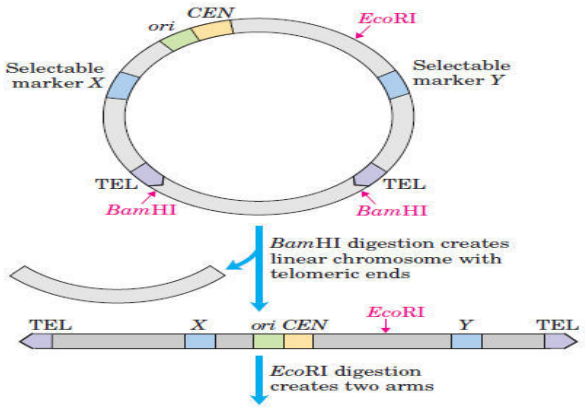
Cosmids, phasmids and other vectors

Yeast artificial chromosomes (YACs)

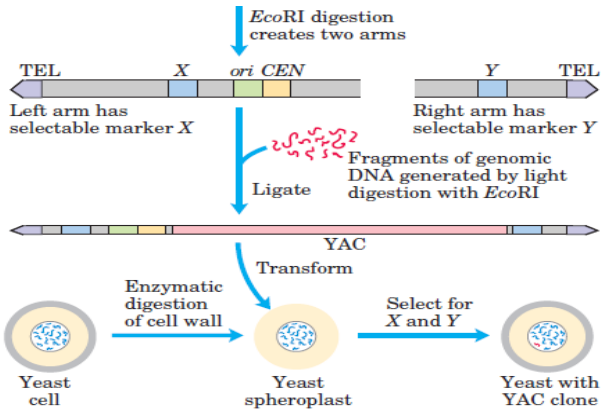
- These vectors replicate in yeast like normal chromosome but they have sites where very large DNA fragments can be inserted

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Construction of yeast artificial chromosome



Construction of yeast artificial chromosome

Cosmids, phasmids and other vectors

Yeast artificial chromosomes (YACs)

- YACs with inserts more than 150,000 bp are nearly as stable as normal cellular chromosomes

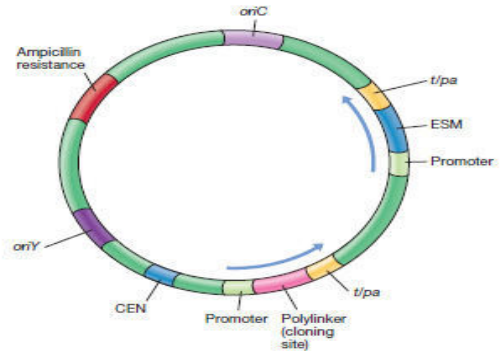
Cosmids, phasmids and other vectors

Shuttle and Expression vectors

- Vectors that can replicate and are stably maintained in two or more unrelated host organisms are called shuttle vectors

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Genetic map of a shuttle vector used in yeast

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Cosmids, phasmids and other vectors

Shuttle and Expression vectors

- To control the expression of cloned genes expression vectors are used

Cosmids, phasmids and other vectors

Comparison of different cloning vectors-summary

- Different cloning vectors can carry variable sizes of target DNA into host organisms

Cosmids, phasmids and other vectors

Table. Maximum DNA insert possible with different cloning vectors

Vector	Host	Insert size
pBR322	*E. coli*	10-15 kb
ʎ phage	*E. coli*	2-25 kb
ʎ Cosmids	*E. coli*	35-45 kb
P1 phage	*E. coli*	70-100 kb
PACs	*E. coli*	100-300 kb
BACs	*E. coli*	≤ 300 kb
YACs	*S. cerevisiae*	200-2000 kb

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Cloning strategies

OVERVIEW OF CLONING STRATEGIES

- Any cell based cloning procedure has four essential parts

i). A method of generating DNA fragment for cloning

ii). A reaction that inserts that fragment into the chosen cloning vectors

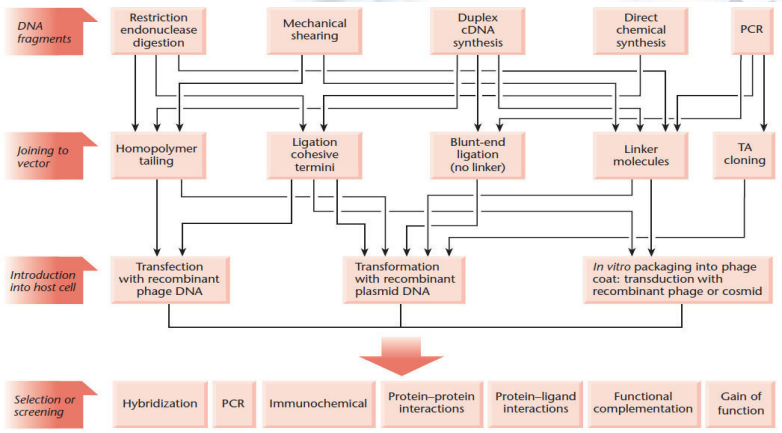
Cloning strategies

Introduction

iii). A means for introducing that recombinant vector into a host cell wherein it is replicated

iv). A method of selecting recipient cells that have acquired the recombinant

Cloning strategies



Generalized overview of cloning strategies

Cloning strategies

Introduction

- There are two major strategies for isolating sequences from complex sources

i). Cell based cloning strategy, is to divide the source DNA into manageable fragments and clone everything

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Cloning strategies

Introduction

ii). The second strategy is to selectively amplify the target sequence directly from the source DNA by using PCR

Cloning genomic DNA

Genomic DNA libraries

- Producing representative genomic libraries in λ cloning vector

Cloning genomic DNA

Table 1. Genome size of selected organisms

Organism	Genome size (kb) (haploid where appropriate)
Escherichia coli	4.0×10^3
Yeast (*Saccharomyces cerevisiae*)	1.35×10^4
Arabidopsis thaliana (higher plant)	1.25×10^5
Tobacco	1.6×10^6
Wheat	5.9×10^6
Zea mays	1.5×10^7
Drosophila melanogaster	1.8×10^5
Mouse	2.3×10^6
Human	2.8×10^6
Xenopus laevis	3.0×10^6

Cloning genomic DNA

Genomic DNA libraries

- Cloning of large DNA fragments of 20 kb in λ replacement vectors, fewer clones are required for nearly complete library

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Cloning genomic DNA

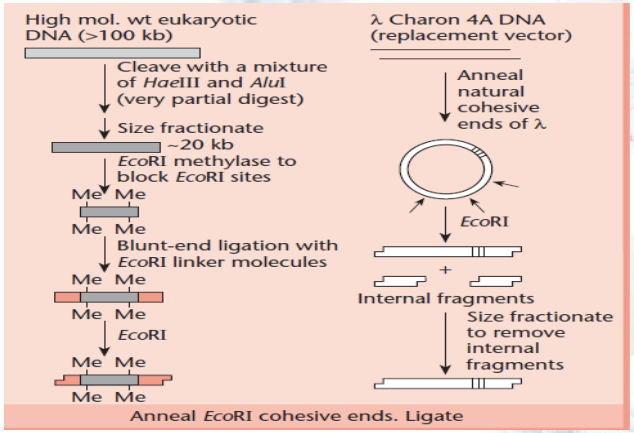
Genomic DNA libraries

- How can appropriately sized random fragments be produced?

i). Average fragment size can be controlled by mechanical shearing

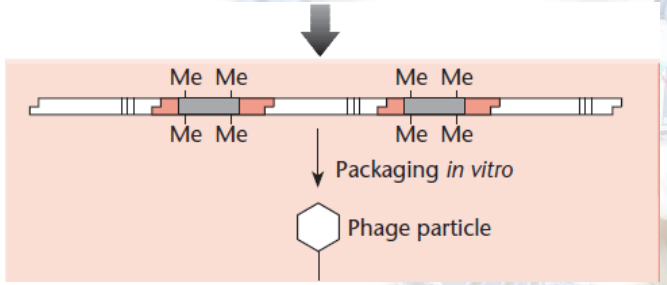
ii). The more commonly used procedure is restriction endonucleases

Cloning genomic DNA



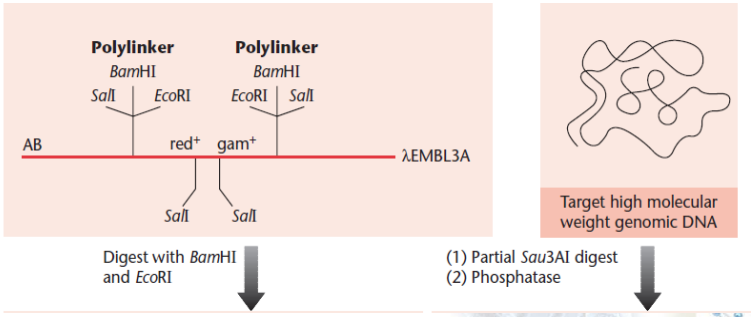
Maniatis' strategy for producing a representative gene library

Cloning genomic DNA



Maniatis' strategy for producing a representative gene library

Cloning genomic DNA

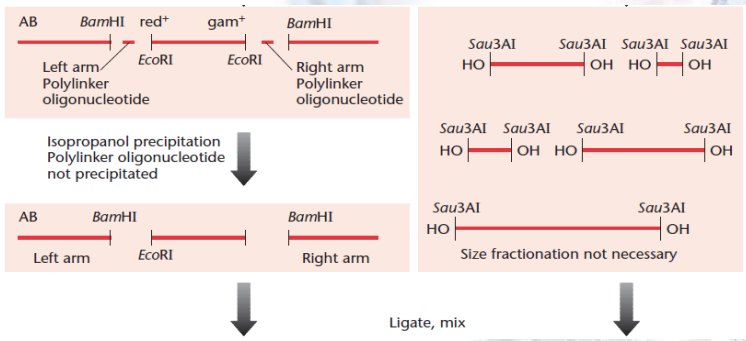


Creation of genomic DNA library using the phage-λ vector EMBL3A

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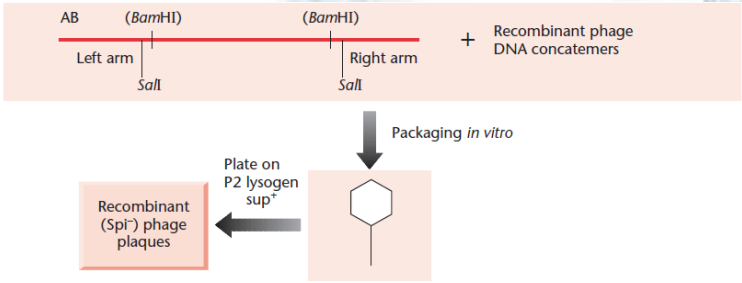
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Cloning genomic DNA



Creation of genomic DNA library using the phage-ʎ vector EMBL3A

Cloning genomic DNA



Creation of genomic DNA library using the phage-ʎ vector EMBL3A

Cloning genomic DNA

Genomic libraries in high-capacity vectors

- In place of phage-ʎ derivatives, a number of high capacity cloning vectors such as cosmids, bacterial artificial chromosomes (BACs) and yeast artificial chromosomes (YACs) are available for construction of genomic libraries

Cloning genomic DNA

PCR as an alternative to genomic DNA cloning

- PCR with specific primers could be used to isolate genes directly from genomic DNA, obviating the need for the production of genomic libraries

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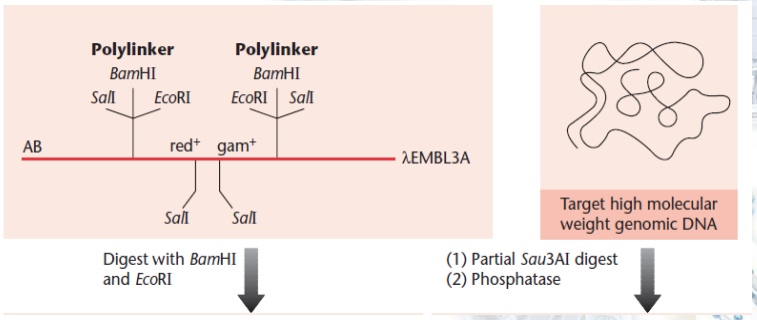
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Cloning genomic DNA

λEMBL vectors for genomic library construction

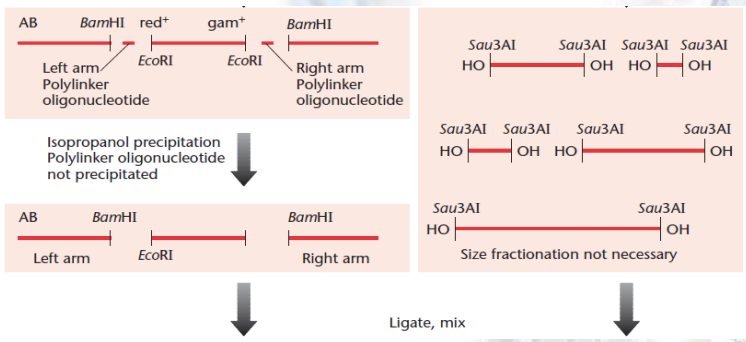
- High molecular weight genomic DNA is digested with *Sau*3AI and subsequently ligated in λEMBL vector digested with *Bam*H1

Cloning genomic DNA



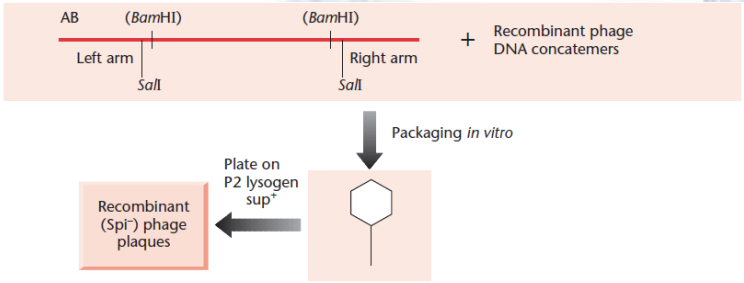
Creation of genomic DNA library using the phage-λ vector EMBL3A

Cloning genomic DNA



Creation of genomic DNA library using the phage-λ vector EMBL3A

Cloning genomic DNA



Creation of genomic DNA library using the phage-λ vector EMBL3A

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Cloning genomic DNA

Genomic libraries in high-capacity vectors

- In place of phage-λ derivatives, a number of high capacity cloning vectors such as cosmids, bacterial artificial chromosomes (BACs) and yeast artificial chromosomes (YACs) are available for construction of genomic libraries

Cloning genomic DNA

PCR as an alternative to genomic DNA cloning

- PCR with specific primers could be used to isolate genes directly from genomic DNA, obviating the need for the production of genomic libraries

cDNA cloning

Properties of cDNA

- cDNA is prepared by reverse-transcribing cellular RNA
- cDNA lack introns and other non-coding sequences present in the corresponding genomic DNA
- Introns are rare in bacteria but occur in genes of higher eukaryotes

cDNA cloning

cDNA libraries

- cDNA library is a combination of cloned cDNA fragments inserted into a collection of host cells
- cDNA library is representative of the RNA population from which it was derived

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cDNA cloning

Table. Abundance classes of typical mRNA populations

Source	Number of different mRNAs	Abundance (molecules/cell)
Mouse liver cytoplasmic poly(A)+	9	12 000
	700	300
	11 500	15
Chick oviduct polysomal poly(A)+	1	100 000
	7	4 000
	12 500	5

References: mouse (Young *et al.* 1976); chick oviduct (Axel *et al.* 1976).

cDNA cloning

cDNA libraries

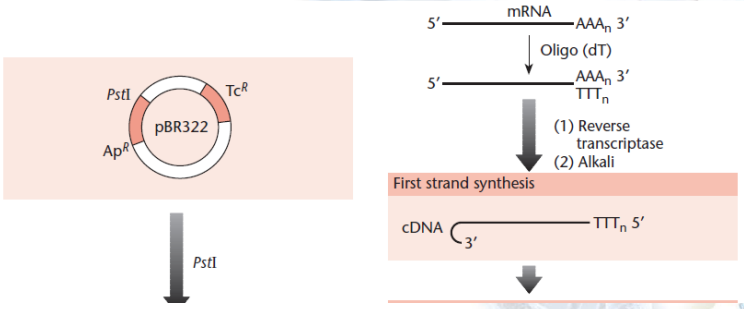
- Phage-λ vectors for cDNA cloning and expression
- λgt10 and λgt11 vectors

cDNA cloning

Preparation of cDNA for library construction

- The cDNA synthesis reaction
- Development of cDNA cloning strategies

cDNA cloning

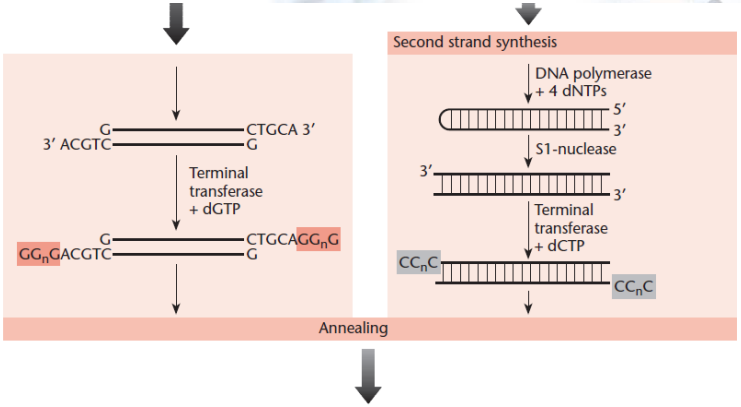


An early cDNA cloning strategy

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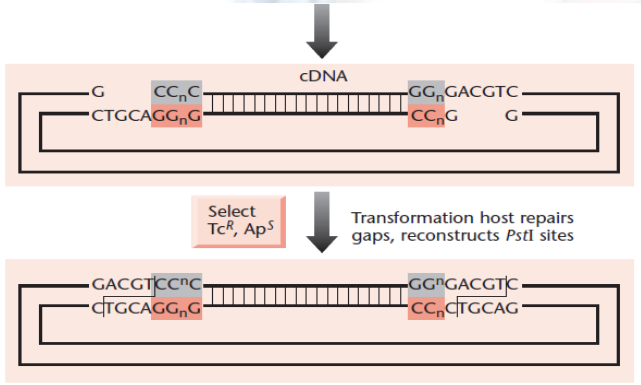
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cDNA cloning



An early cDNA cloning strategy

cDNA cloning



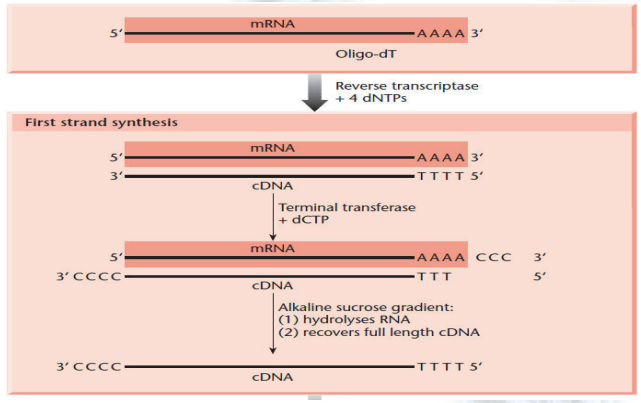
An early cDNA cloning strategy

cDNA cloning

Improved methods for cDNA cloning

- A serious disadvantage of the hairpin method is that cleavage with S1 nuclease results in the loss of sequences at the 5′ end of the clone
- This strategy has therefore been superseded with other methods

cDNA cloning

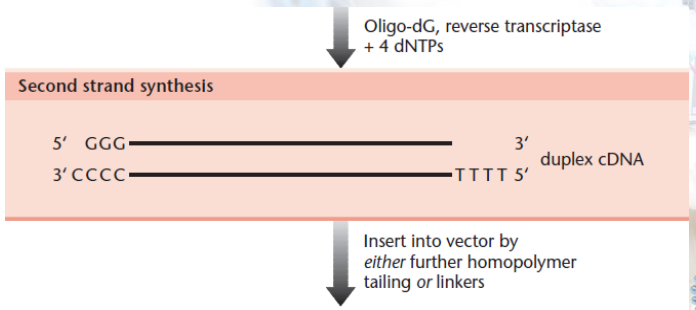


Improved method for cDNA cloning (Land et al. 1981)

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cDNA cloning



Improved method for cDNA cloning (Land et al. 1981)

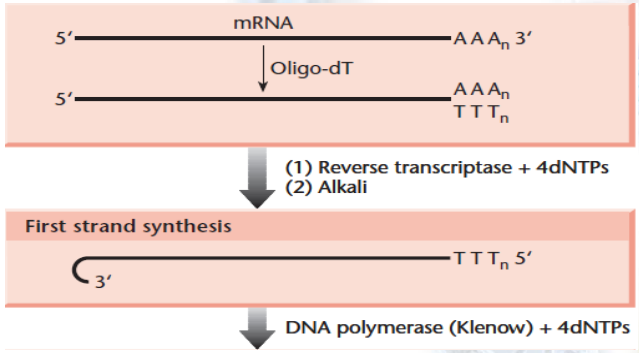
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cDNA cloning

Improved methods for cDNA cloning

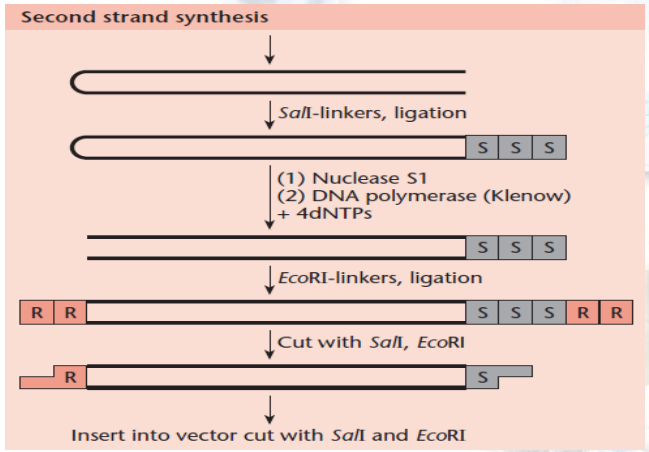
- For cDNA expression libraries, it is advantageous if the cDNA can be inserted into the vector in the correct orientation
- This can be achieved by self-priming method by adding linker molecule to double stranded cDNA

cDNA cloning



Method for direction cDNA cloning

cDNA cloning



Method for direction cDNA cloning

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cDNA cloning

PCR as an alternative for cDNA cloning

- Reverse transcription followed by the PCR (RT-PCR) leads to the amplification of RNA sequences in cDNA form
- RT-PCR is a sensitive means for detecting, quantifying and cloning specific cDNA molecules

Screening strategies

- Major screening strategies involve

1). Genetic methods
2). Sequence-dependent screening
3). Screening expression libraries

Genetic methods

Selection for the presence of vector

- All useful vector molecules carry a selectable genetic marker or property
- Plasmid and cosmid vectors carry drug resistance or nutritional marker
- In phage vectors, plaque formation is itself the selected property

Sequence-dependent screening

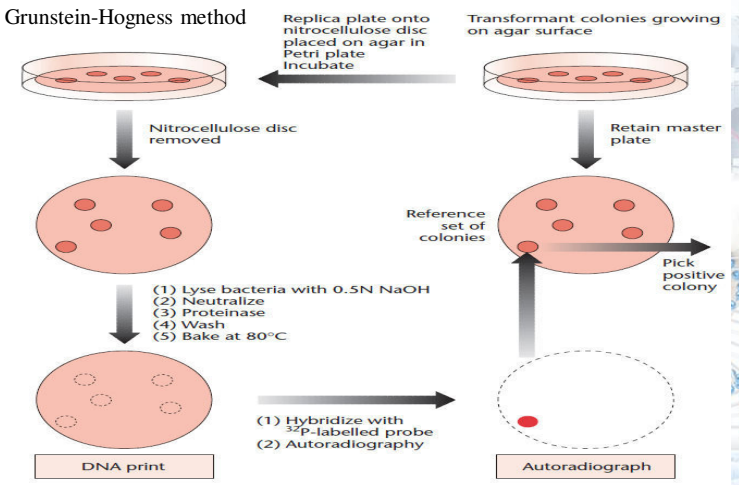
Screening by hybridization

- Nucleic acid hybridization is the most commonly used method of library screening
- Grunstein and Hogness (1975) developed a screening procedure to detect DNA sequences in transformed colonies by *in situ* hybridization

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Sequence-dependent screening



Sequence-dependent screening

Screening by hybridization

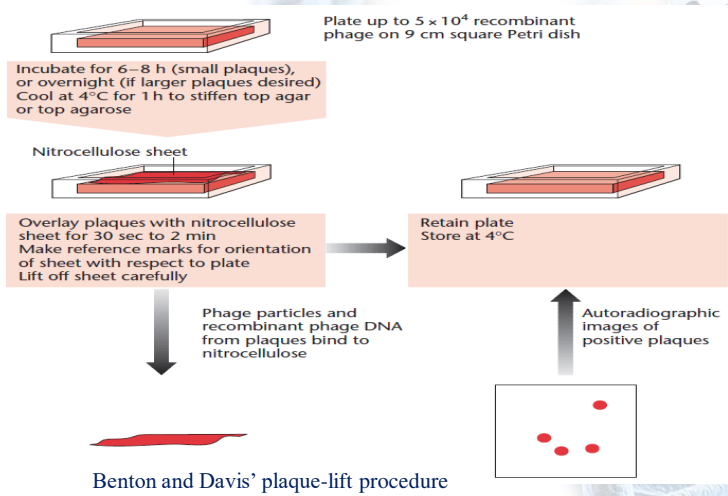
- The results of the hybridization can be monitored by autoradiography

Sequence-dependent screening

Benton and Davis' plaque lift procedure

- Benton and Davis (1977) devised a method called plaque lift, in which nitrocellulose filter is applied to the upper surface of agar plates, making direct contact between plaques and filter

Sequence-dependent screening



Benton and Davis' plaque-lift procedure

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Sequence-dependent screening

Benton and Davis' plaque lift procedure

- Number of alternative labeling methods are available that avoid the use of radioactivity. It included digoxigenin or biotin

Sequence-dependent screening

Probe design

- A great advantage of hybridization for library screening is that it is extremely versatile
- Conditions can be used in which hybridization is very stringent, so that only sequences identical to the probe are identified

Sequence-dependent screening

Chromosome walking

- Walking along the chromosome is a term used to describe an approach which allows the isolation of gene sequences whose function is quite unknown but whose genetic location is known
- A cloned genomic fragment must be found as a starting point for the walk

Sequence-dependent screening

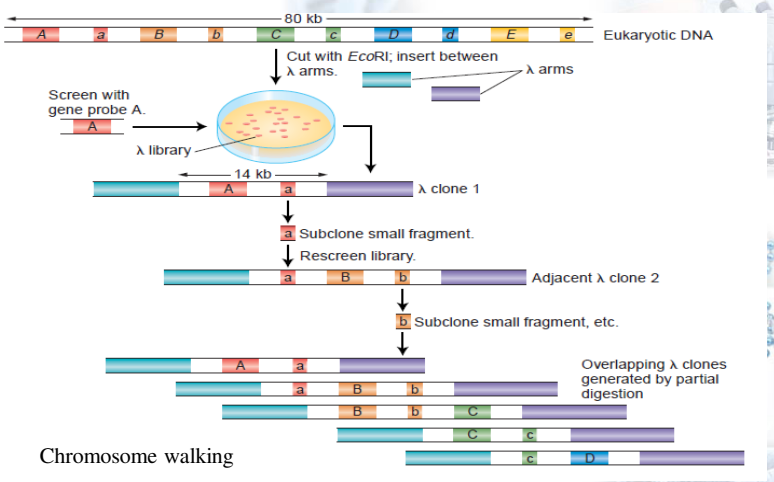
Chromosome walking

- A cloned genomic fragment must be found as a starting point for the walk
- In human genome, the starting point may be a restriction fragment length polymorphism (RFLP) sequence that is closely linked to disease locus

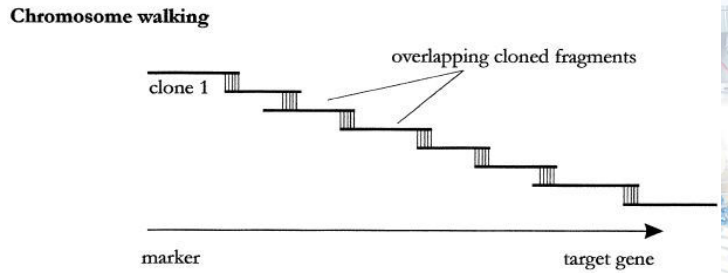
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Sequence-dependent screening



Sequence-dependent screening



Sequence-dependent screening

Chromosome walking

- One drawback to this method is the requirement that each DNA segment used is not repeated elsewhere in the genome

Sequence-dependent screening

Chromosome jumping

- In chromosome jumping, the DNA of interest is identified, cut into fragments with restriction enzymes and circularized
- It brings together DNA sequence that were originally located a considerable distance apart in the genome

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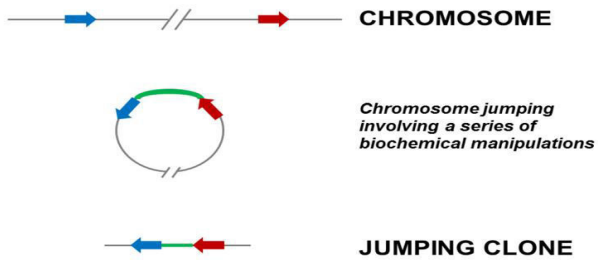
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Sequence-dependent screening

Chromosome jumping

- These cloned DNAs from the closure sites make up a jumping library

Screening strategies



Sequence-dependent screening

Chromosome jumping

- One of the application of chromosome walking and jumping is in the cloning of human cystic fibrosis gene

Sequence-dependent screening

Screening by PCR

- The PCR is widely used to isolate specific DNA sequences from uncloned genomic DNA or cDNA but it also a useful technique for library screening
- To isolate specific clone, PCR is carried out with gene-specific primers that flank a unique sequence in the target

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Sequence-dependent screening

Screening by PCR

- There are several applications where the use of degenerate primes is favorable
- A degenerate primer is a mixture of primers, all of similar sequences but with variations at one or more positions

Screening expression libraries

Expression cloning

- If DNA library is established using expression vectors, each individual clone can be expressed to yield a polypeptide
- Expression libraries are useful because they allow a range of alternative techniques to be employed

Screening expression libraries

Immunochemical screening

- Immunochemical screening involves the use of antibodies that specifically recognize antigenic determinants on the polypeptides synthesized by a target clone
- The molecular target for recognition is generally an epitope

Screening expression libraries

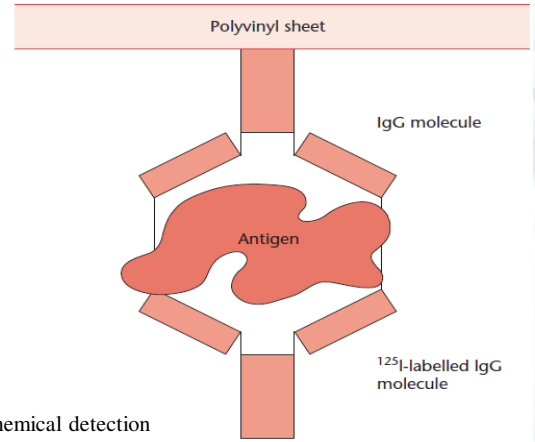
Immunochemical screening

- The method of Broome and Gilbert (1978) exploited the fact that antibodies adsorb very strongly to certain types of plastic such as polyvinyl and that IgG antibodies can be readily labelled with ^{125}I by iodination *in vitro*

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Screening expression libraries



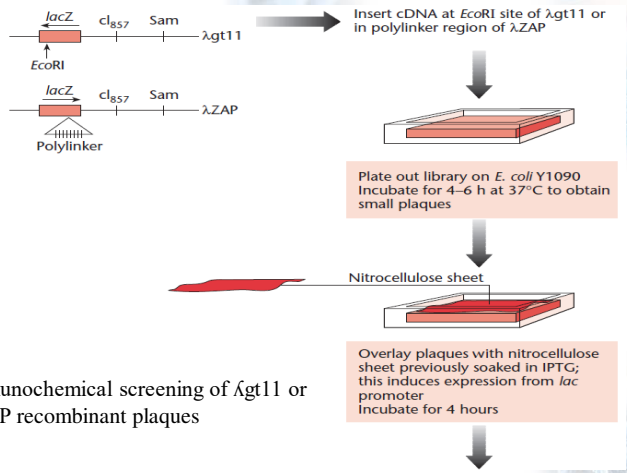
Immunochemical detection method of Broome and Gilbert

Screening expression libraries

Immunochemical screening of λgt11

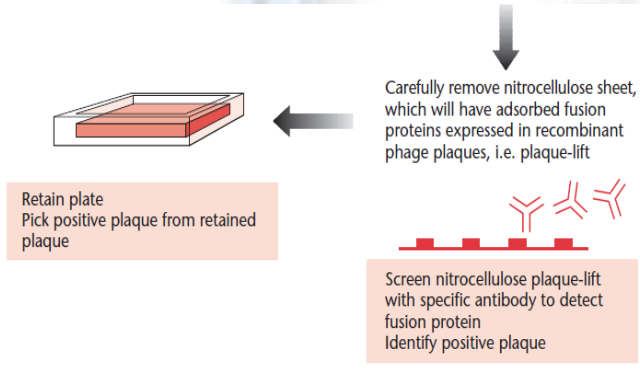
- It is much more convenient to use bacteriophage-λ insertion vectors, because these have a higher capacity and the efficiency of *in vitro* packaging allows large numbers of recombinants to be prepared and screened

Screening expression libraries



Immunochemical screening of λgt11 or λZAP recombinant plaques

Screening expression libraries



Immunochemical screening of λgt11 or λZAP recombinant plaques

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Screening expression libraries

Immunochemical screening of λgt11

- The original detection method using iodinated antibodies has been superseded by more convenient methods using non-isotopic labels, which are also more sensitive and have a lower background of non-specific signal

Screening expression libraries

South-western and north-western blotting

- It involves the screening and isolation of clones expressing sequence specific DNA or RNA-binding proteins
- Screening is carried out without using an antibody, by incubating the membranes with radiolabelled double stranded DNA probe

Screening expression libraries

Screening by functional complementation

- Functional complementation is the process by which a particular DNA sequence compensates for a missing function in a mutant cell and thus restores the wild type phenotype

Screening expression libraries

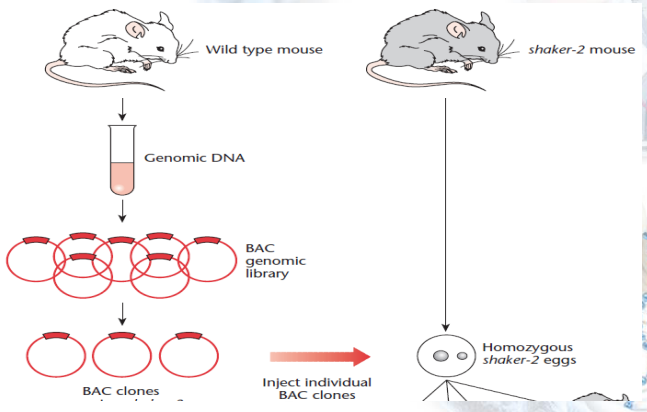
Screening by functional complementation

- Functional complementation is also possible in transgenic animals and plants
- It has been used for complementation in transgenic mice to isolate the *Shaker-2* gene

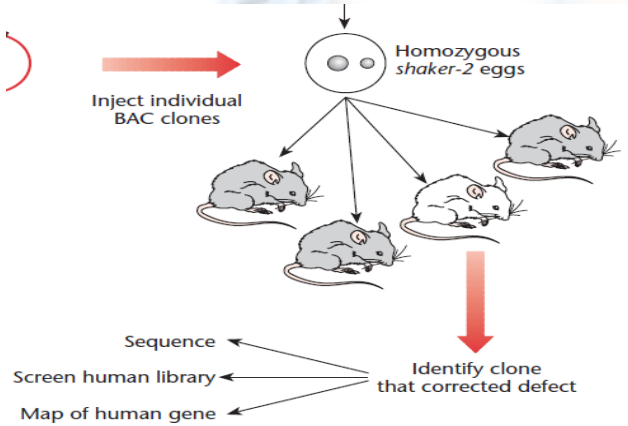
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Screening expression libraries



Screening expression libraries



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Expression in *E. coli* of cloned DNA

Requirement for expression in *E. coli*

- Synthesis of functional protein depends upon transcription of appropriate gene, efficient translation of mRNA and in many cases, post-translational processing and compartmentalization of nascent polypeptide

Expression in *E. coli* of cloned DNA

Secretion of proteins

- Gram-negative bacteria such as *E. coli* have a complex wall-membrane structure comprising an inner, cytoplasmic membrane separated from an outer membrane by a cell wall and periplasmic space

Expression in *E. coli* of cloned DNA

Protein trafficking

- The bacterial inner membrane, periplasmic space and outer membrane all contain proteins not found in the cytoplasm
- All of these proteins are first synthesized in the cytoplasm but somehow find their way out

Expression in *E. coli* of cloned DNA

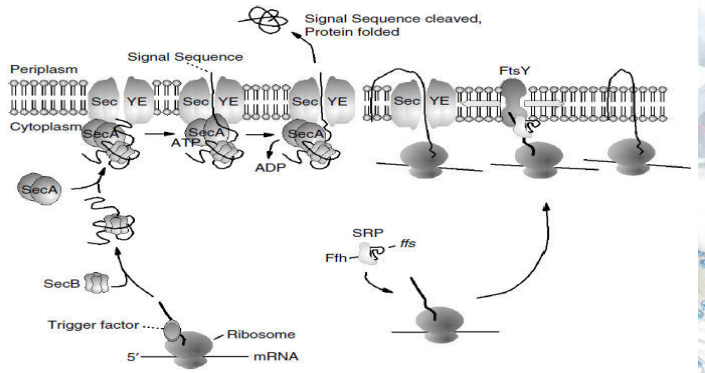


Figure. General secretion pathway

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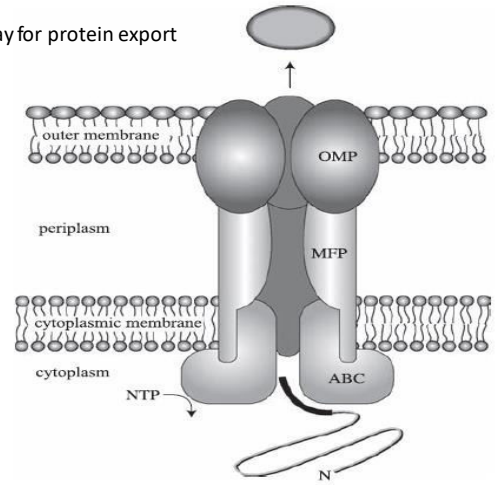
Expression in *E. coli* of cloned DNA

Protein trafficking

- There are actually several general mechanisms of protein export in bacteria

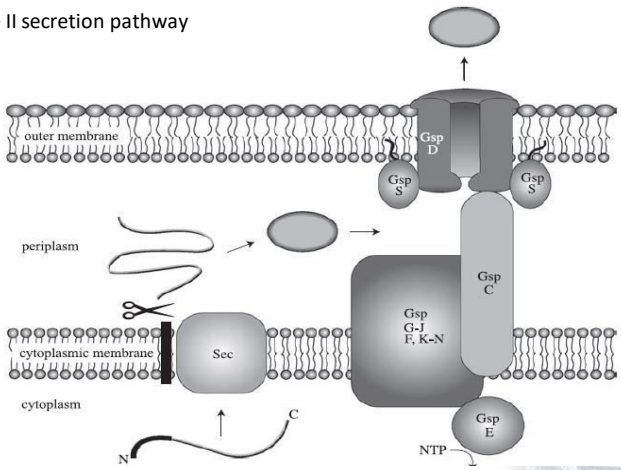
Expression in *E. coli* of cloned DNA

ABC pathway for protein export



Expression in *E. coli* of cloned DNA

Type II secretion pathway



Expression in *E. coli* of cloned DNA

Protein trafficking

- Each system transport a particular type of proteins

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Expression in *E. coli* of cloned DNA

Stability of foreign proteins in *E. coli*

- Various strategies have been developed to cope with the instability of foreign proteins in *E. coli*
- In the case of somatostatin, degradation was prevented by producing a fused protein consisting of somatostatin and β-galactosidase

Maximizing the expression of cloned gene

Constructing the optimal promoter

- Large number of promoters for *E. coli* have been analysed
- Many promoters has led to the formulation of a consensus sequence which consists of the -35 region (5′-TTGACA-) and -10 region or Pribnow box (5′-TATAAT-)

Maximizing the expression of cloned gene

	-35 Region		-10 Region
		1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17	
CONSENSUS	•••TTGACA	•••••••••• •••••••••	TATAAT••
lac	GGCTTTACA	CTTTATGCTTCCGGCTCG	TATATTGT
trp	CTGTTGACA	ATTAATCAT CGAACTAG	TTAACTAG
λP_L	GTGTTGACA	TAAATACCA CTGGCGGT	GATACTGA
rec A	CACTTGATA	CTGTATGAA GCATACAG	TATAATTG
tacI	CTGTTGACA	ATTAATCAT CGGCTCG	TATAATGT
tacII	CTGTTGACA	ATTAATCAT CGAACTAG	TTTAATGT

Figure. The base sequence of the -10 and -35 regions for natural and hybrid promoters

Maximizing the expression of cloned gene

Constructing the optimal promoter

- Expression from a strong promoter can represent 20-40% of cloned gene product of toral cell protein

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Maximizing the expression of cloned gene

Optimizing translation initiation

- Complementarity of Shine-Dalgarno (S-D) sequence with 16s rRNA can affect the rate of translation
- Composition of triplet immediately preceding the AUG start codon also affects the efficiency of translation

Maximizing the expression of cloned gene

Stability of mRNA and codon choice

- The rate of synthesis of a particular protein will depend on the steady-state of mRNA in the cell
- Degradation of mRNA usually proceeds by a combination of endonuclease and 3′-exonuclease attack

Maximizing the expression of cloned gene

The effect of plasmid copy number

- The number of ribosomes in a cell far exceeds any one class of mRNA
- One way of increasing the expression of a cloned gene is to increase the number of the corresponding transcript

Maximizing the expression of cloned gene

Plasmid stability

- Having maximized the expression of a particular gene it is important to consider what effects this will have on the bacterium harbouring the recombinant plasmid
- The loss of plasmids due to defective partitioning is called segregative instability

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Maximizing the expression of cloned gene

Plasmid stability

- Structural instability of plasmids may arise by deletions or rearrangements of DNA

Maximizing the expression of cloned gene

Host cell physiology can affect the level of expression

- Factors which will be important include the choice of nutrients and environmental parameters

Expression in *E. coli* of cloned DNA

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CONSENSUS	•••TTGACA	•••••••••• •••••••••	TATAAT••
lac	GGCTTTACA	CTTTATGCTTCCGGCTCG	TATATTGT
trp	CTGTTGACA	ATTAATCAT CGAACTAG	TTAACTAG
λP_L	GTGTTGACA	TAAATACCA CTGGCGGT	GATACTGA
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Analysing DNA sequences

DNA sequencing: Benefits and Applications

- DNA sequence information is a prerequisite for planning any substantial manipulation of the DNA
- Information is useful to various fields i.e. molecular and evolutionary biology, metagenomics, medicine, forensics etc.

Analysing DNA sequences

DNA sequencing: Benefits and Applications

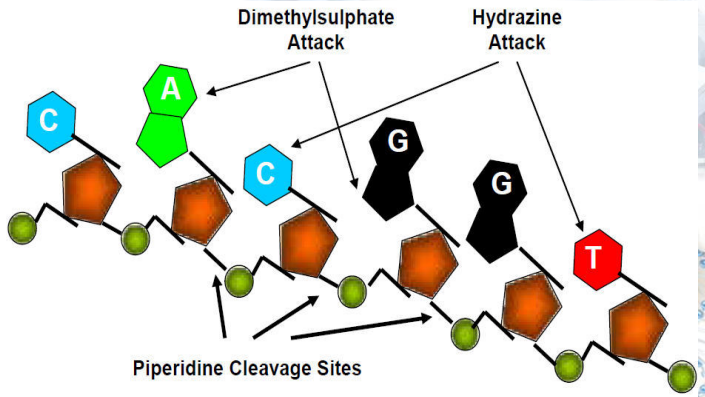
- Techniques for DNA sequencing became available in the late 1970s

Analysing DNA sequences

Maxam-Gilbert method

- This method for DNA sequencing makes use of chemical reagents to bring about base-specific cleavage of the DNA

Analysing DNA sequences



Chemical targets in Maxam and Gilbert DNA sequencing strategy

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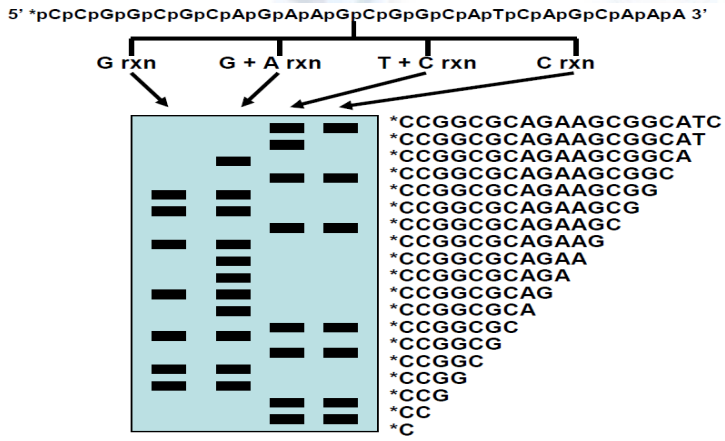
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Analysing DNA sequences

Table. Reagents for Maxam and Gilbert DNA sequencing

Base specificity	Base reaction	Altered base removal	Strand cleavage
G	Dimethylsulphate	Pepridine	Pepridine
G+A	Acid	Acid-catalysed depurination	Pepridine
T+C	Hydrazine	Pepridine	Pepridine
C	Hydrazing+NaCl	Pepridine	Pepridine
A > C	NaOH	Pepridine	Pepridine

Analysing DNA sequences



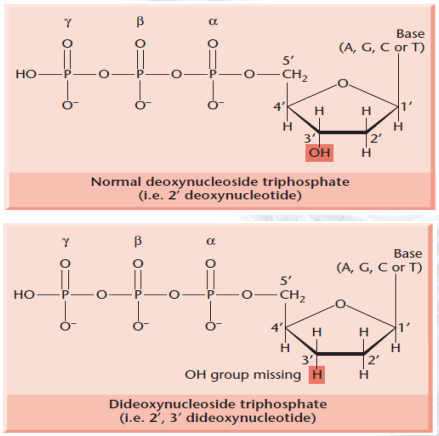
The Maxam and Gilbert mannual sequencing scheme

Analysing DNA sequences

Chain termination or dideoxy procedure

- This method was developed by Sanger et al. (1977)
- It capitalizes on the ability of DNA polymerase to incorporate dideoxynucleoside triphosphates as substrate

Analysing DNA sequences



Dideoxynucleoside triphosphate act as chain terminator

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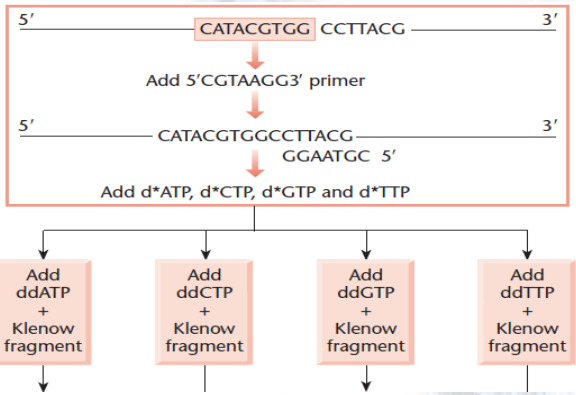
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Analysing DNA sequences

Chain termination or dideoxy procedure

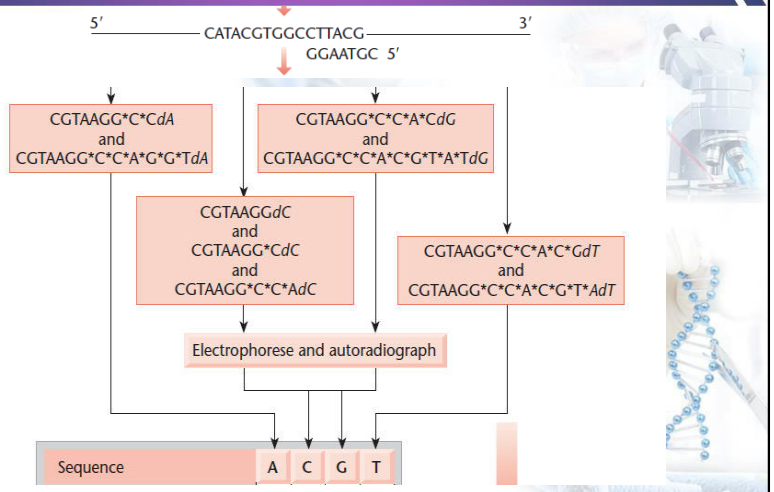
- DNA synthesis is carried out in the presence of the four deoxynucleoside triphospahtes, one or more of which is labelled with ^{32}P and in four separate incubation mixes

Analysing DNA sequences

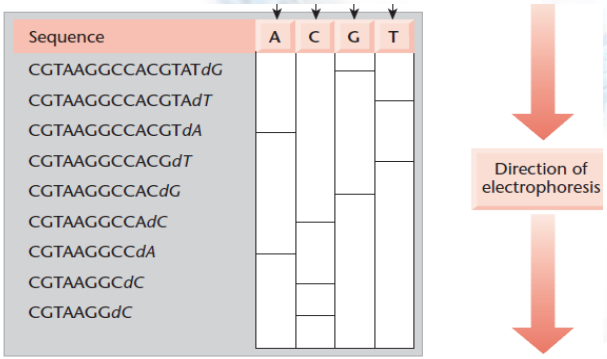


DNA sequencing by chain terminator method

Analysing DNA sequences



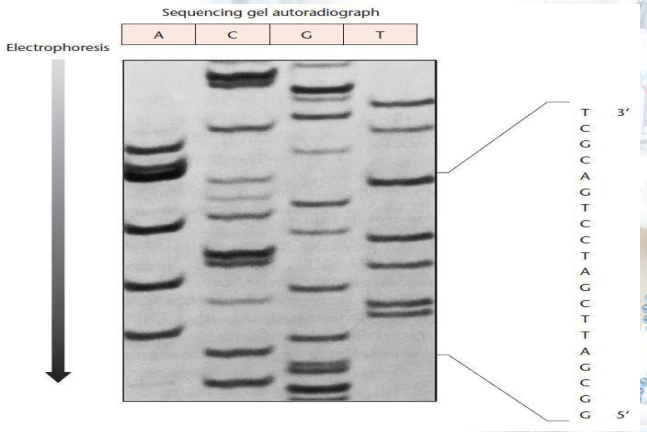
Analysing DNA sequences



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Analysing DNA sequences



Analysing DNA sequences

Modifications of chain terminator sequencing

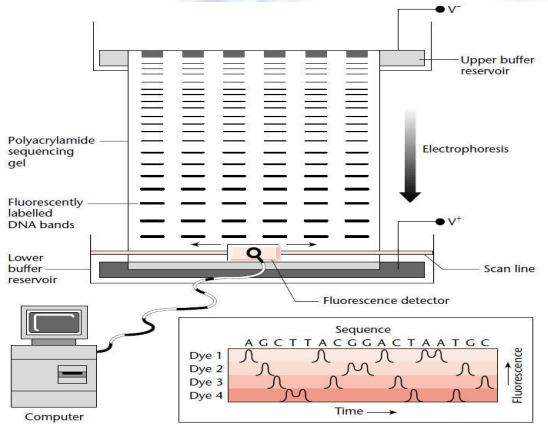
- Improvements to original Sanger's method have been made by replacing the Klenow fragment of *E. coli* DNA polymerase I
- The combination of chain terminator and M13 vectors to produce single stranded DNA is very powerful

Analysing DNA sequences

Automated DNA sequencing

- To automate the process, it is desirable to acquire sequence data in real time by detecting the DNA bands within the gel during the electrophoretic separation

Analysing DNA sequences

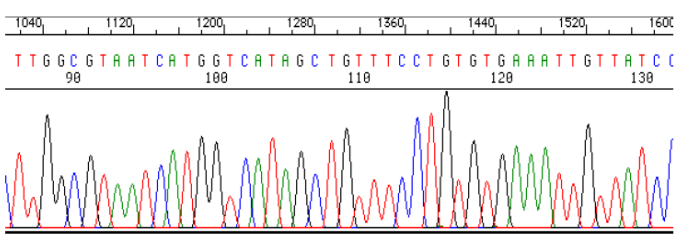


Block diagram of an automated DNA sequencer

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Analysing DNA sequences



Sequence electropherogram

Analysing DNA sequences

Automated DNA sequencing

- Automated DNA sequencing offers a number of advantages

Analysing DNA sequences

Sequencing accuracy

- Sequences that were read beyond 400 bp contained an average of 3.2% error, while those less than 400 bp had 2.8% error

Analysing DNA sequences

DNA sequence databases

- Since the development of current DNA sequencing technology large amount of sequence data has accumulated that is maintained in 3 data bases

i). National center for Biotechnology Information
ii). DNA Databank of Japan
iii). European Bioinformatic institute-UK

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Changing genes: site-directed mutagenesis

Mutagenesis

- Mutagenesis is a process to change the genetic information of an organism
- It can occur naturally, or as a result of exposure to mutagens or induced experimentally in laboratory

Changing genes: site-directed mutagenesis

Mutagenesis

- Three different methods of site-directed mutagenesis have been devised

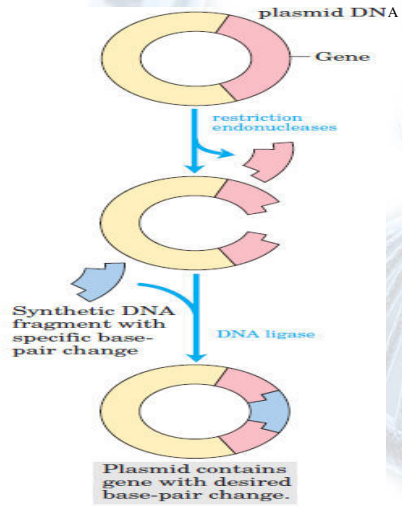
i). Cassette mutagenesis
ii). Primer extension
iii). Procedures based on PCR

Changing genes: site-directed mutagenesis

Cassette mutagenesis

- A synthetic DNA fragment containing the desired mutant sequence is used to replace the corresponding sequence in the wild-type gene

Changing genes: site-directed mutagenesis



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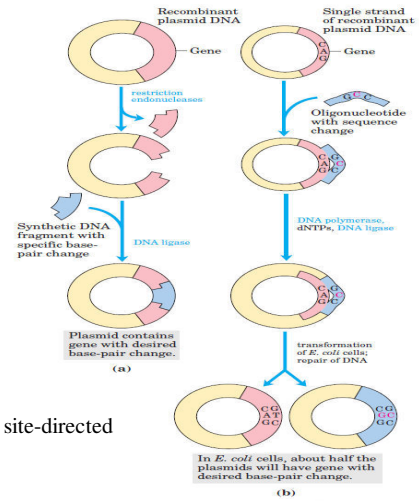
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Changing genes: site-directed mutagenesis

Primer extension: the single primer method

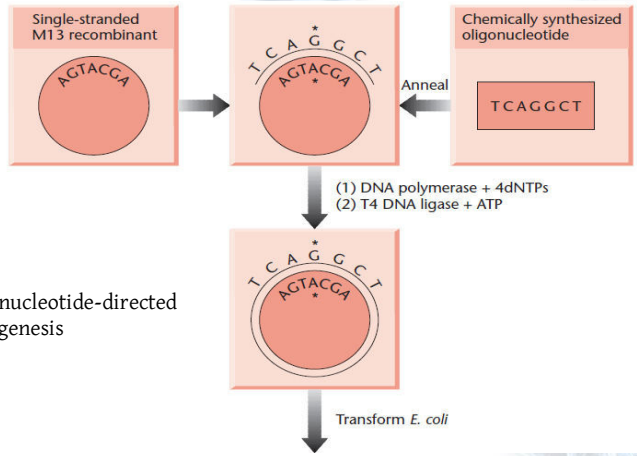
- It involves priming in vitro DNA synthesis with a chemically synthesized oligonucleotide (7-20 bp) that carries a base mismatch with the complementary sequence

Changing genes: site-directed mutagenesis



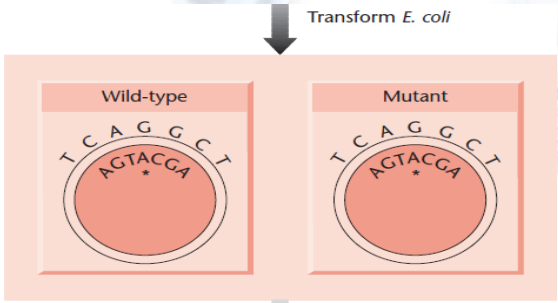
Two approaches to site-directed mutagenesis

Changing genes: site-directed mutagenesis



Oligonucleotide-directed mutagenesis

Changing genes: site-directed mutagenesis



Oligonucleotide-directed mutagenesis

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Changing genes: site-directed mutagenesis

Primer extension: the single primer method

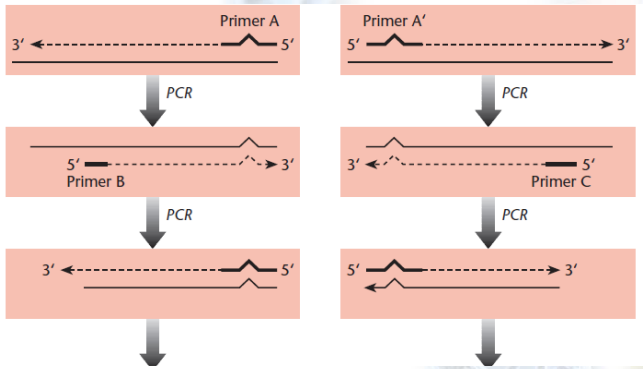
- After *E. coli* transformation, heteroduplex that are formed are either that of original wild type DNA or that containing the mutated base

Changing genes: site-directed mutagenesis

PCR methods for site-directed mutagenesis

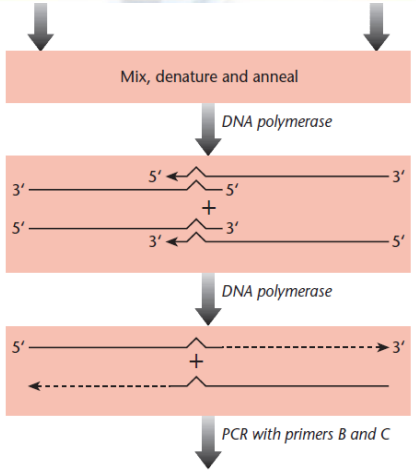
- Single bases mismatched between the amplification primer and the template become incorporated into the template sequence as a result of amplification

Changing genes: site-directed mutagenesis



Site directed mutagenesis by means of the PCR

Changing genes: site-directed mutagenesis



Site directed mutagenesis by means of the PCR

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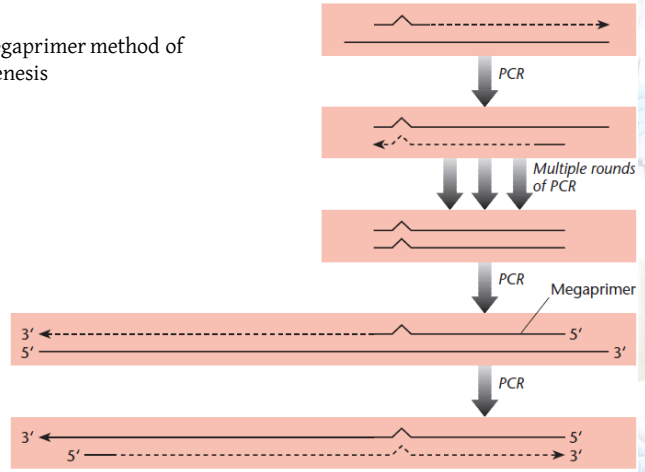
Changing genes: site-directed mutagenesis

PCR methods for site-directed mutagenesis

- In megaprimer method, the product of the first PCR is used for the second PCR

Changing genes: site-directed mutagenesis

The megaprimer method of mutagenesis



Changing genes: site-directed mutagenesis

PCR methods for site-directed mutagenesis

- The advantage of a PCR-based mutagenic protocol is that the desired mutation is obtained with 100% efficiency

Polymerase chain reaction (PCR)

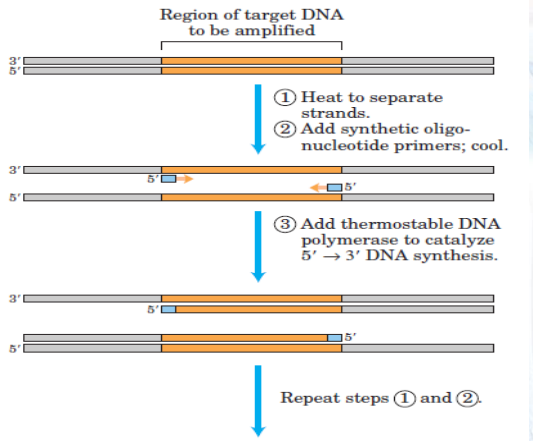
Basic PCR reaction

- PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude, generating millions of copies of a particular DNA sequence

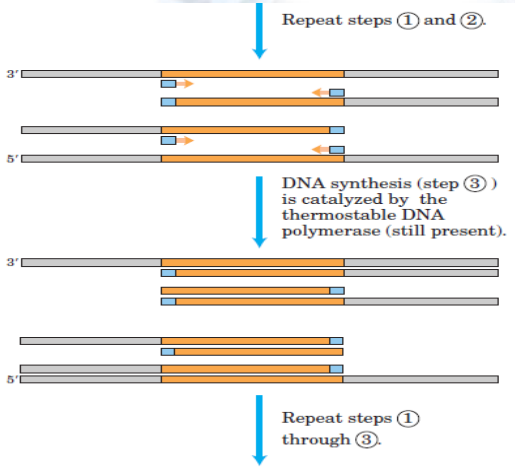
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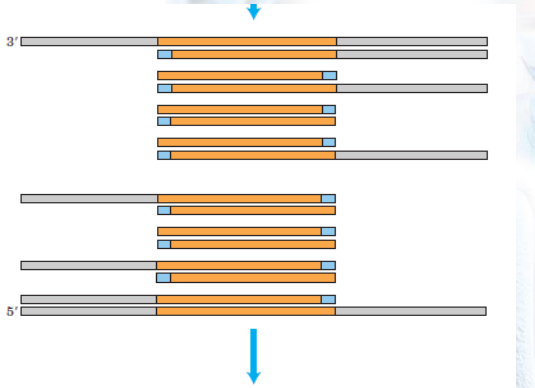
Polymerase chain reaction (PCR)



Polymerase chain reaction (PCR)



Polymerase chain reaction (PCR)



Polymerase chain reaction (PCR)

Cycle number	Number of double-stranded target molecules
1	0
2	0
3	2
4	4
5	8
6	16
7	32
8	64
9	128
10	256
11	512
12	1024
13	2048
14	4096
15	8192
16	16,384
17	32,768
18	65,536
19	131,072
20	262,144
21	524,288
22	1,048,576
23	2,097,152
24	4,194,304
25	8,388,608
26	16,777,216
27	33,554,432
28	67,108,864
29	134,217,728
30	268,435,456

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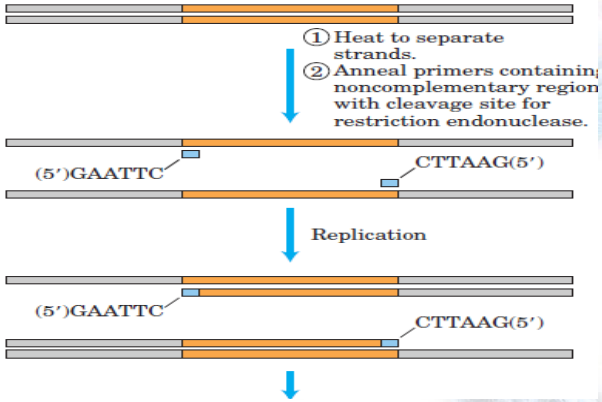
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Polymerase chain reaction (PCR)

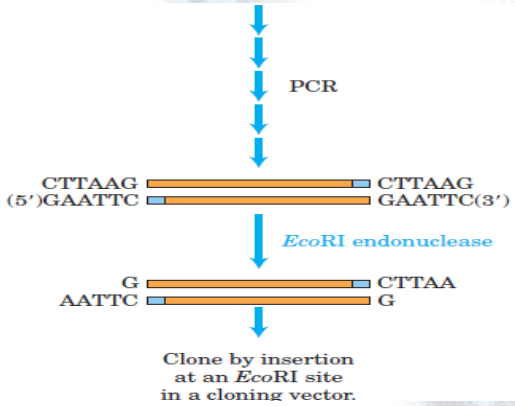
Basic PCR reaction

- PCR can also be used to generate DNA fragments for gene cloning by using primers with specific restriction sites

Polymerase chain reaction (PCR)



Polymerase chain reaction (PCR)



Polymerase chain reaction (PCR)

Basic PCR reaction

- PCR is a technology in molecular biology used to amplify a single or a few copies of a target DNA across several orders of magnitude, generating millions of copies of a particular DNA sequence

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Polymerase chain reaction (PCR)

PCR principles and procedure

- PCR amplifies a specific target region of DNA that may range from a few to several kilo base pairs
- Basic reaction requires several components
- PCR cycles may consists of several steps of denaturation, annealing and extension

Polymerase chain reaction (PCR)

DNA polymerases

- DNA polymerases are enzymes that synthesize DNA molecules by adding nucleotide, the building blocks of DNA

Polymerase chain reaction (PCR)

Table. Comparison of DNA polymerases of E. coli

	DNA polymerase		
	I	II	III
Structural gene*	*polA*	*polB*	*polC* (*dnaE*)
Subunits (number of different types)	1	7	≥10
M_r	103,000	88,000†	791,500
3′→5′ Exonuclease (proofreading)	Yes	Yes	Yes
5′→3′ Exonuclease	Yes	No	No
Polymerization rate (nucleotides/s)	16-20	40	250-1,000
Processivity (nucleotides added before polymerase dissociates)	3-200	1,500	≥500,000

Polymerase chain reaction (PCR)

Taq DNA polymerase

- Taq polymerase is a thermostable DNA polymerase isolated from thermophilic bacterium *Thermus aquaticus* and is used in PCR to amply DNA fragments

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Polymerase chain reaction (PCR)

Table. Sources of thermostable DNA polymerases

DNA polymerase	Source
Tma	*Thermotoga maritima*
Deep Vent™	*Pyrococcus* sp.
Tli	*Thermococcus litoralis*
Pfu	*Pyrococcus furiosus*
Pwo	*Pyrococcus woesi*

Polymerase chain reaction (PCR)

Primers

- Primer is a short strand of oligonucleotide that serves as a starting point for DNA synthesis

Polymerase chain reaction (PCR)

Degenerate primers

- A degenerate primer is actually a mixture of primers, all of similar sequence but with variations at one or more positions

Polymerase chain reaction (PCR)

Table. Amino acids by codon specificity

One Codon	Two Codons	Three Codons	Four Codons	Six Codons
Met (M)	Cys (C)	Ile (I)	Ala (A)	Leu (L)
Trp (W)	Asp (D)		Gly (G)	Arg (R)
	Glu (E)		Pro (P)	Ser (S)
	Phe (F)		Thr (T)	
	His (H)		Val (V)	
	Lys (K)			
	Asn (N)			
	Gln (Q)			
	Tyr (Y)			

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Polymerase chain reaction (PCR)

Table. Standard nucleotide coding system

Symbol	Base represented
A	A
C	C
G	G
T	T
R	A, G
Y	C, T
M	A, C
K	G, T
H	A, C, T
B	C, G, T
V	A, C, G
N	A, C, G, T

Polymerase chain reaction (PCR)

Degenerate primers

- A degenerate primer is actually a mixture of primers, all of similar sequence but with variations at one or more positions

Polymerase chain reaction (PCR)

Types of PCR

- Recently, variants have been developed from basic PCR method to improve performance and specificity and to amplify other molecules such as RNA

Polymerase chain reaction (PCR)

RT-PCR

- Reverse transcriptase PCR (RT-PCR) was designed to amplify RNA sequences (mainly mRNA) through synthesis of cDNA by reverse transcriptase

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Polymerase chain reaction (PCR)

RT-PCR

- Various strategies can be adopted for first-strand cDNA synthesis

Polymerase chain reaction (PCR)

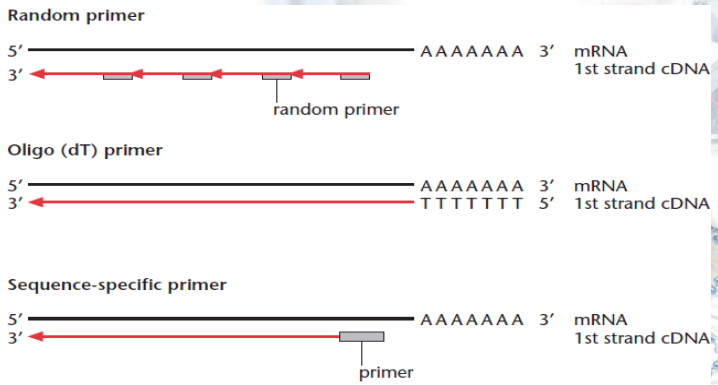


Fig. Strategies for synthesis of first-strand cDNA

Polymerase chain reaction (PCR)

RT-PCR

- Various strategies can be adopted for first-strand cDNA synthesis

Polymerase chain reaction (PCR)

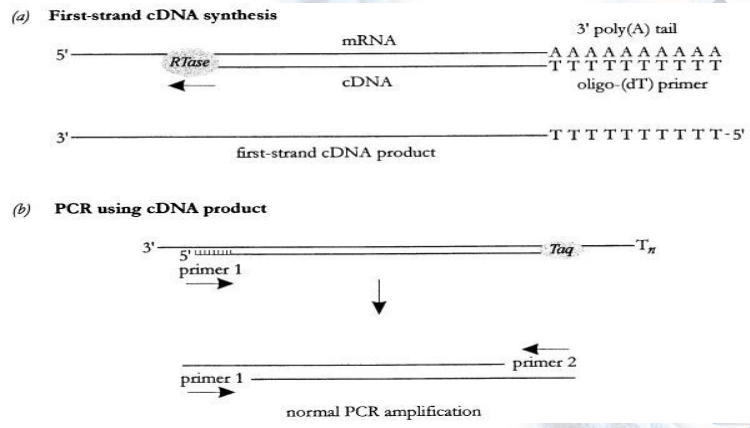


Fig. RT-PCR

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Polymerase chain reaction (PCR)

Competitor RT-PCR

- One use of RT-PCR is in determining the amount of mRNA in a sample (competitor RT-PCR)

Polymerase chain reaction (PCR)

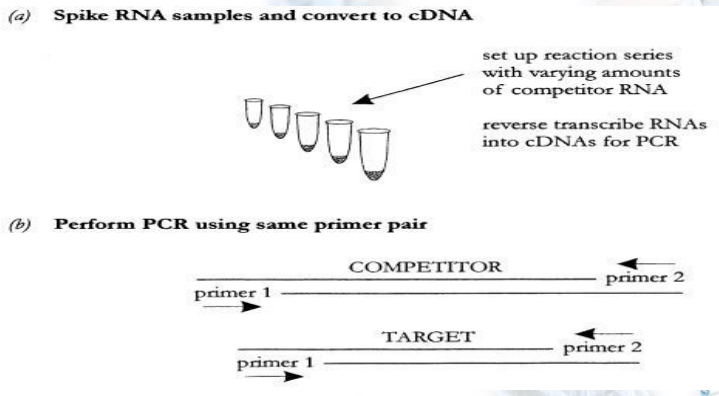


Fig. Competitor RT-PCR

Polymerase chain reaction (PCR)

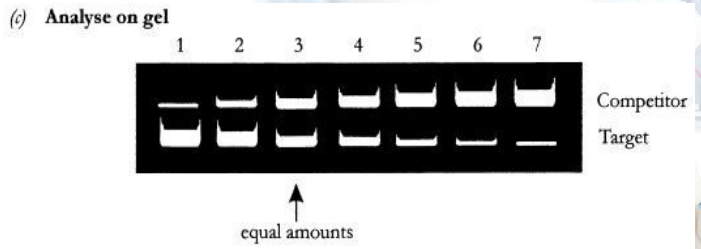


Fig. Competitor RT-PCR

Polymerase chain reaction (PCR)

RT-PCR

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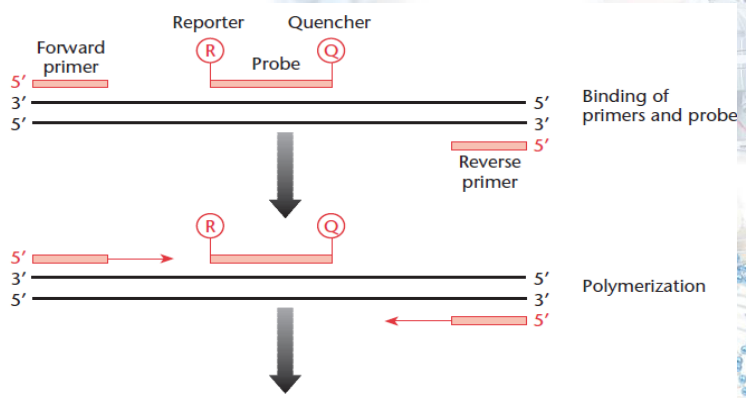
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Polymerase chain reaction (PCR)

Real-time quantitative PCR (qPCR)

- qPCR is used to measure the quantity of a target sequence
- It is used to measure starting amounts of DNA, cDNA, or RNA

Polymerase chain reaction (PCR)



Polymerase chain reaction (PCR)

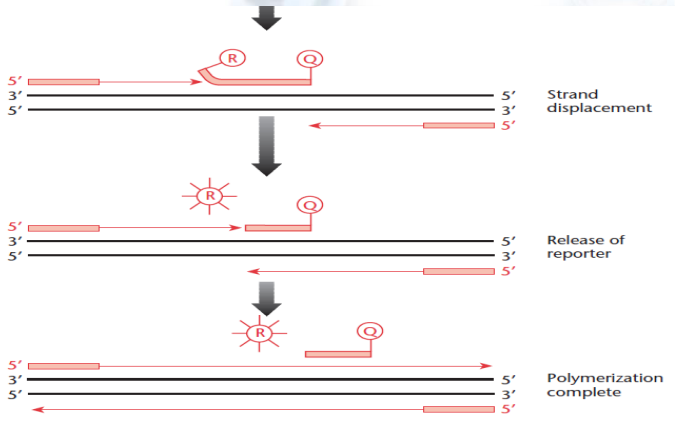


Fig. Real-time quantitative PCR

Polymerase chain reaction (PCR)

Real-time quantitative PCR (qPCR)

- qPCR is used to measure the quantity of a target sequence
- It is used to measure starting amounts of DNA, cDNA, or RNA

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Polymerase chain reaction (PCR)

Nested PCR

- In nested PCR, two sets of primers are used in two successive PCRs
- First set of primers allows a first amplification
- Product of first PCR is subjected to a second PCR using the second set of primers

Polymerase chain reaction (PCR)

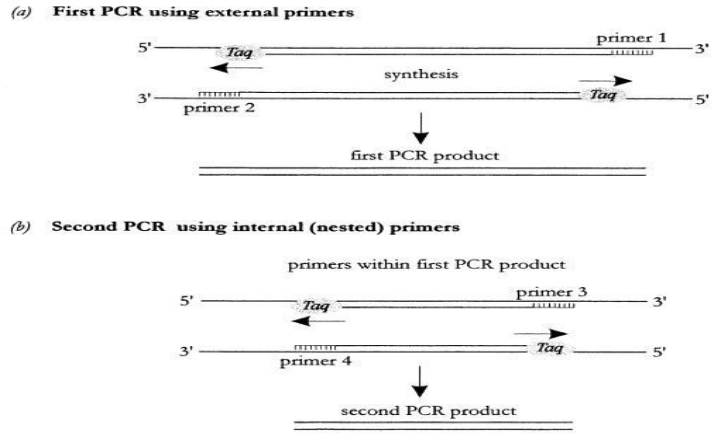


Fig. Nested PCR

Polymerase chain reaction (PCR)

Nested PCR

- In nested PCR, two sets of primers are used in two successive PCRs
- First set of primers allows a first amplification
- Product of first PCR is subjected to a second PCR using the second set of primers

Polymerase chain reaction (PCR)

Inverse PCR

- Inverse PCR is used to identify the flanking sequences around genomic inserts

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Polymerase chain reaction (PCR)

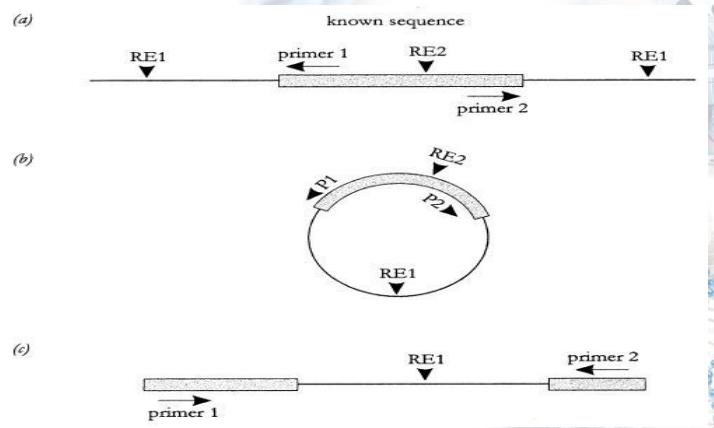


Fig. Inverse PCR

Polymerase chain reaction (PCR)

Multiplex PCR

- Multiplex PCR is used to amplify several different DNA sequences simultaneously

Polymerase chain reaction (PCR)

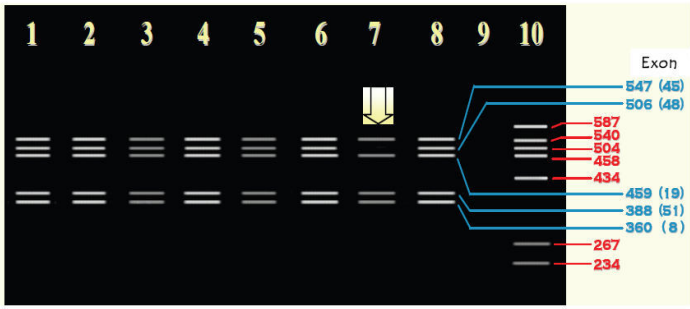


Fig. Results of multiplex PCR in a patient with Duchenne Muscular Dystrophy Inverse PCR

Hernandex-Rodriguez et al., 2000;
Hernandex-Rodriguez & Restrepo, 2002
(www.intechopen.com)

Polymerase chain reaction (PCR)

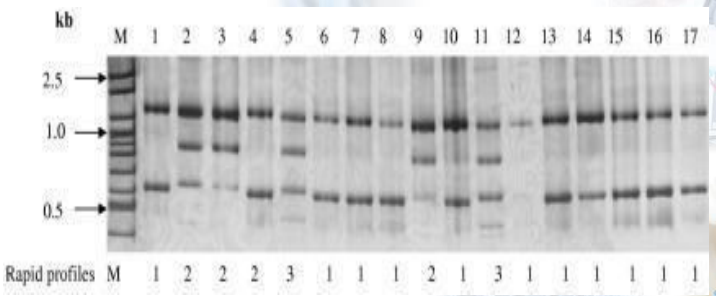
RAPD

- RAPD-Random amplification of polymorphic DNA is a type of PCR reaction but the segments of DNA that are amplified are random

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Polymerase chain reaction (PCR)



Polymerase chain reaction (PCR)

RAPD

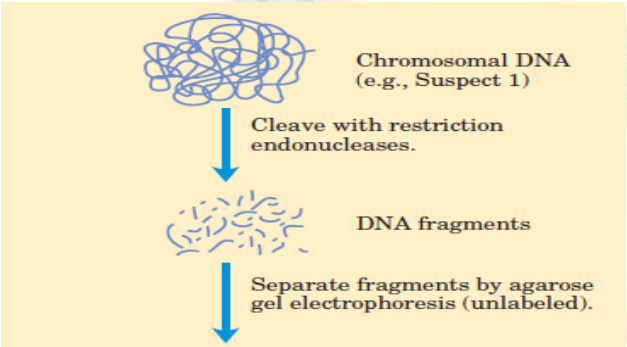
- RAPD-Random amplification of polymorphic DNA is a type of PCR reaction but the segments of DNA that are amplified are random

Polymerase chain reaction (PCR)

RFLP

- RFLP-Restriction fragment length polymorphism exploits variations in homologous DNA sequences
- In RFLP, DNA sample is digested by restriction enzymes and separated by gel electrophoresis

Polymerase chain reaction (PCR)

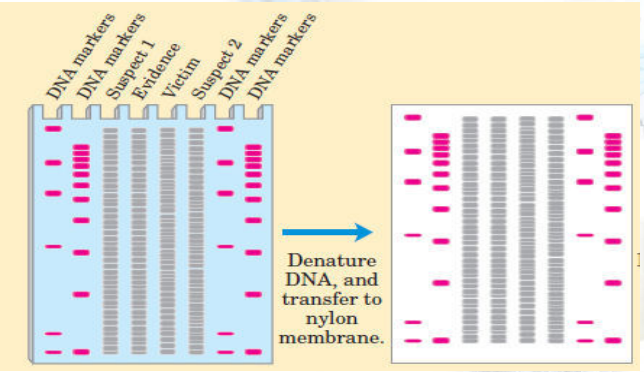


DNA fingerprinting

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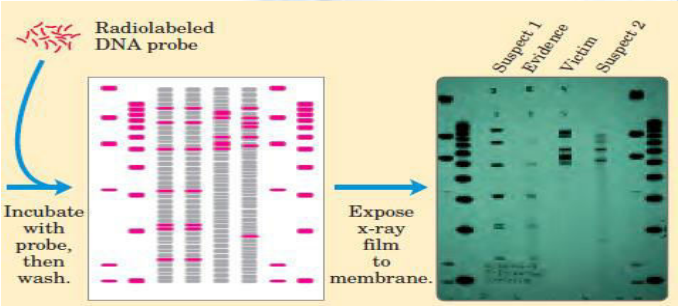
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Polymerase chain reaction (PCR)



DNA fingerprinting

Polymerase chain reaction (PCR)



DNA fingerprinting

Polymerase chain reaction (PCR)

RFLP

- RFLP-Restriction fragment length polymorphism that exploits variations in homologous DNA sequences
- In RFLP, DNA sample is digested by restriction enzymes and separated by gel electrophoresis

Polymerase chain reaction (PCR)

AFLP-PCR

- AFLPs-Amplified fragment length polymorphisms are differences in restriction fragments length caused by SNPs or INDELS
- AFLP is a PCR based tool used in genetic engineering.

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Polymerase chain reaction (PCR)

Applications of PCR

- PCR has widespread applications in various fields of life sciences including genetic engineering, medical, forensic, agriculture, environment etc.

Polymerase chain reaction (PCR)

PCR-Gene cloning and expression

- PCR has been used in gene cloning and screening of genomic libraries

Polymerase chain reaction (PCR)

PCR-Medicine

- PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis
- Molecular tools have also allowed to perform prenatal genetic diagnosis

Polymerase chain reaction (PCR)

PCR-Forensic sciences

- Forensic science is the application of scientific procedures to solve criminal and legal matters
- Molecular methods are used to established the filiations of a person or to obtain evidence from minimal samples of saliva, semen or other tissues

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Polymerase chain reaction (PCR)

PCR-DNA profiling

- DNA profiling or DNA fingerprinting is a forensic technique used to identify individuals by characteristics of their DNA

Polymerase chain reaction (PCR)

PCR-Agricultural sciences and environment

- PCR has also facilitated research in detection of pathogens in plants, animals and environment

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Polymerase chain reaction (PCR)

PCR-Molecular paleontology

- Molecular paleontology refers to the recovery and analysis of DNA and protein from ancient human, animal and plant remains

Mapping and sequencing genomes

Genome mapping

- Assigning of a specific gene to particular region of a chromosome and determining the location of and relative distances between genes on the chromosome
- Types of maps:
  1. Linkage maps
  2. Physical maps

Mapping and sequencing genomes

Markers for genome mapping

- Any identifiable feature of the genome can serve as a marker in mapping
- The landmarks on genome map might include short DNA sequences, regulatory sites and genes themselves

Mapping and sequencing genomes

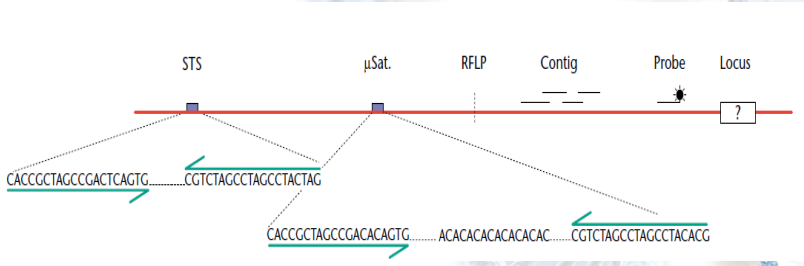


Figure. Markers used in genome mapping
(Paul H Dear, 2001, Encyclopedia of life sciences)

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Methods for genome mapping

Genetic linkage mapping

- Genetic linkage maps illustrate the order of genes or genetic markers on a chromosome and the relative distances between those genes

Methods for genome mapping

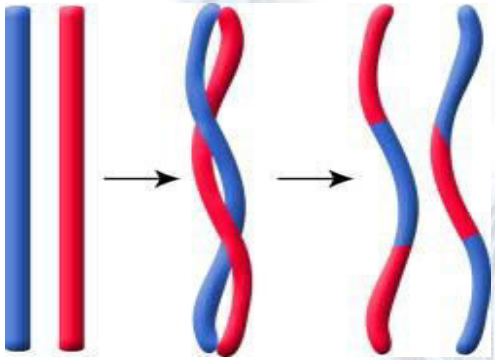


Figure. Crossing over in chromosomes
Source: Genomic News Network (GNN)

Methods for genome mapping

Genetic linkage mapping

- Genetic linkage maps illustrate the order of genes or genetic markers on a chromosome and the relative distances between those genes

Methods for genome mapping

Physical mapping

- Physical maps give the DNA base pair distances from one landmark to another

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Methods for genome mapping

Physical versus linkage maps

- Chromosomal maps of bacteria has been obtained by combining the mapping techniques of interrupted mating, recombination mapping, transformation and transduction

Methods for genome mapping

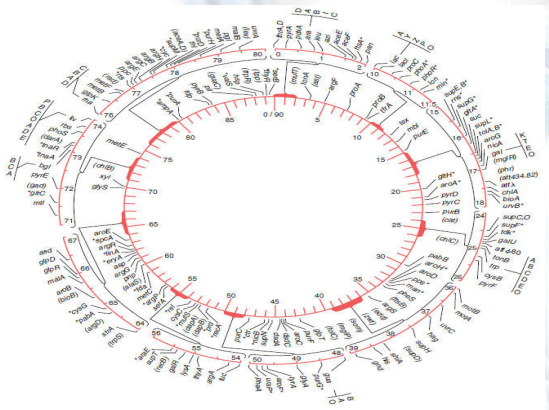


Figure. The 1963 map *of E. coli*

Methods for genome mapping

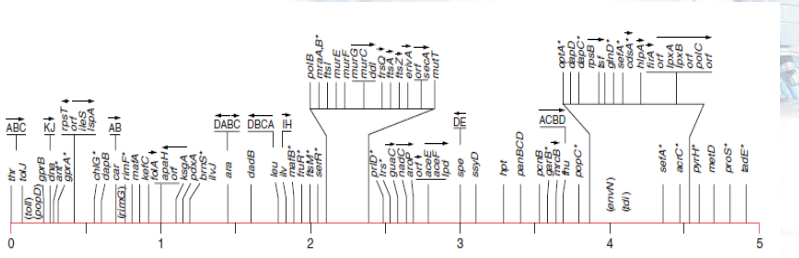


Figure. Linear scale drawing of a 5-minute section of the 100-minute 1990 *E. coli* **linkage map**

Methods for genome mapping

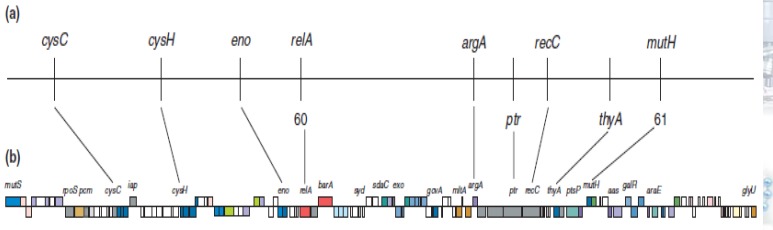


Figure. Correlation of the genetic and physical maps

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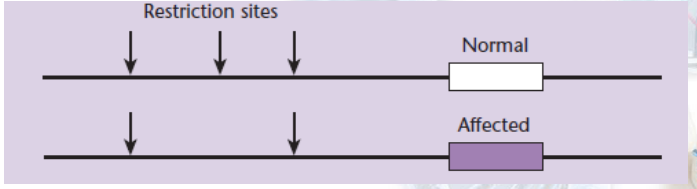
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Methods for genome mapping

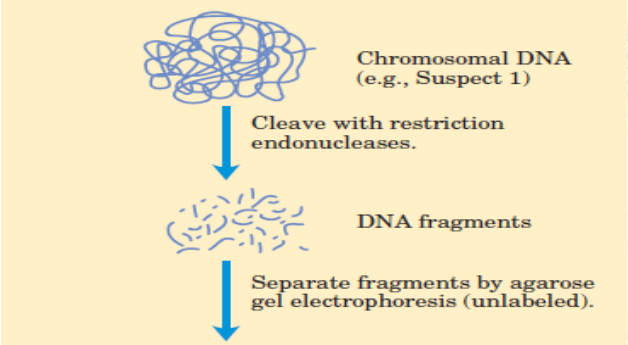
The use of RFLPs in physical maps

- RFLP-Restriction fragment length polymorphism exploits variations in homologous DNA sequences
- Botstein et al. (1980) were the first to recognize that DNA probes that target RFLPs can be used for mapping

Methods for genome mapping

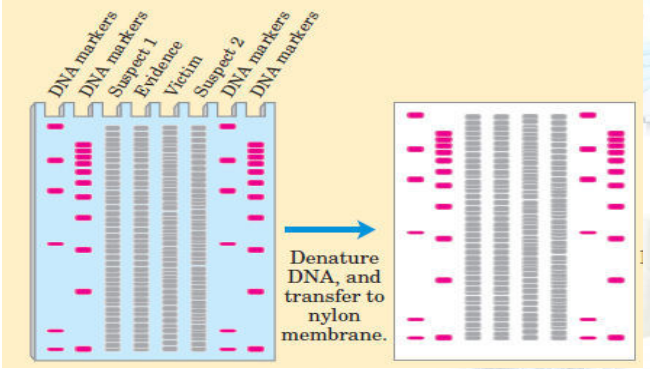


Polymerase chain reaction (PCR)



DNA fingerprinting

Polymerase chain reaction (PCR)



DNA fingerprinting

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Methods for genome mapping

STS in physical maps

- Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use southern blotting

Methods for genome mapping



Figure. Examples of a sequence-tagged site (STS)

Methods for genome mapping

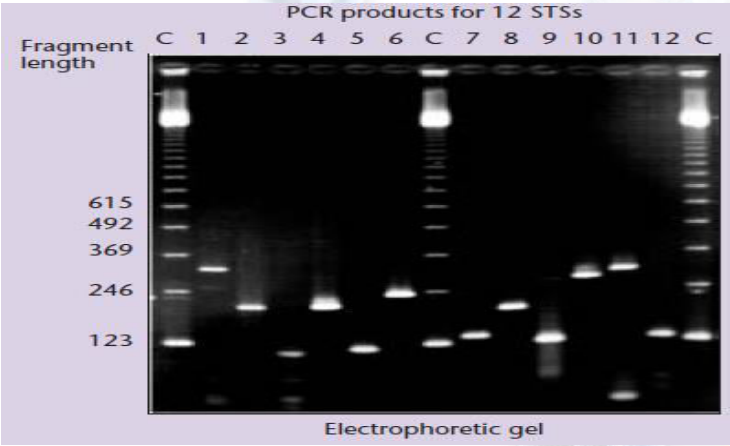


Figure. Confirmation that an STS is a unique sequence on the genome

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STS in physical maps

- Sequence-tagged sites (STS) are more convenient markers than RFLPs because they do not use southern blotting

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Methods for genome mapping

SNPs as physical markers

- Single nucleotide polymorphisms (SNPs) are single base-pair positions in genomic DNA at which different sequence alternatives (alleles) exist in a population
- SNPs probably are the most important sequence markers for physical mapping of genomes

Methods for genome mapping

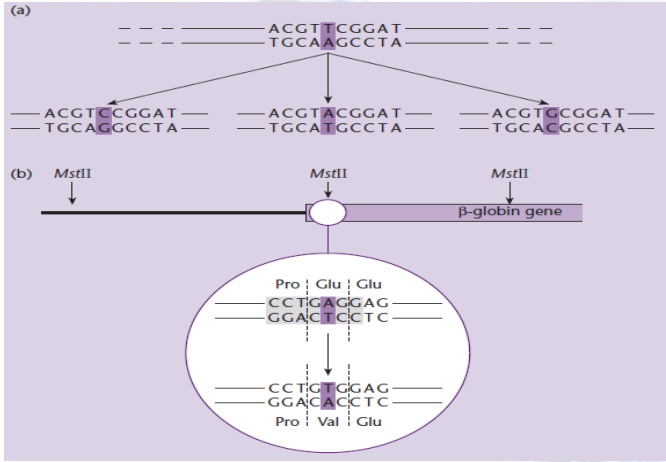


Figure. Examples of single nucleotide polymorphisms

Methods for genome mapping

SNPs as physical markers

- Different methods to detect SNPs have been developed that based on enzymatic, electrophoretic, solid phase or chromatographic analysis

Methods for genome mapping

Polymorphic DNA detection in the absence of sequence information

- Polymorphic DNA can be detected by amplification in the absence of the target DNA sequence information used to generate STSs

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Methods for genome mapping

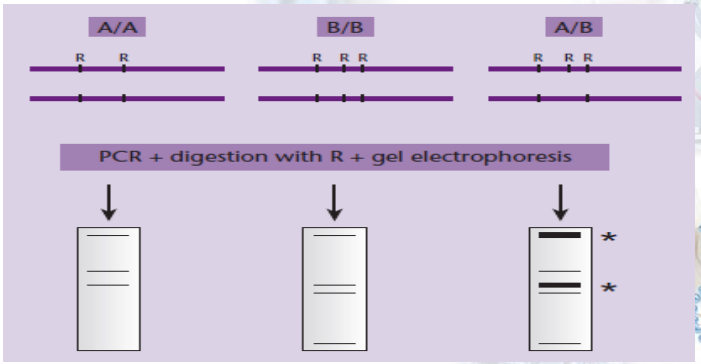


Figure. Generation and visualization of CAPS markers

Methods for genome mapping

AFLPs detection in the absence of sequence information

- Amplified fragment length polymorphisms (AFLP) is a diagnostic fingerprinting technique that detects genomic restriction fragments and in that respect resembles the RFLP technique

Methods for genome mapping

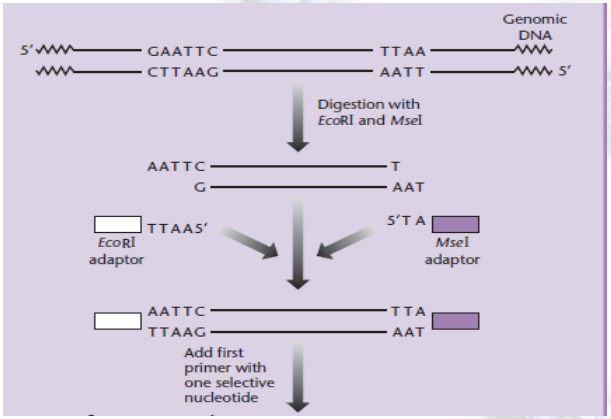


Figure. Principle of the AFLP method

Methods for genome mapping

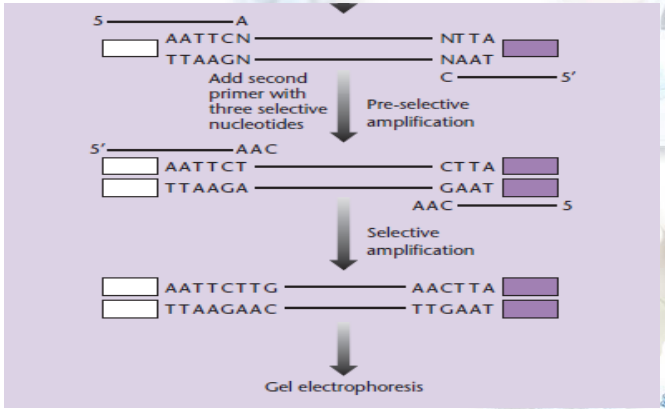


Figure. Principle of the AFLP method

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Methods for genome mapping

AFLPs detection in the absence of sequence information

- Amplified fragment length polymorphisms (AFLP) is a diagnostic fingerprinting technique that detects genomic restriction fragments and in that respect resembles the RFLP technique

Methods for genome mapping

Fluorescence in situ hybridization (FISH)

- FISH is a cytogenetic method that used fluorescent probes that bind to only those parts of the chromosomes with a high degree of sequence complementarity

Methods for genome mapping

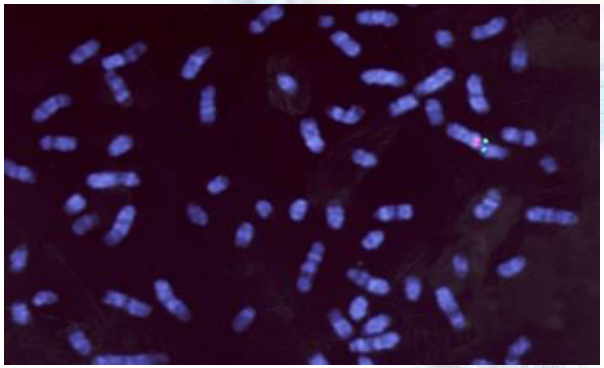


Figure. Dual-color fluorescence in situ hybridization

Methods for genome mapping

Radiation hybrid (RH) mapping

- RH mapping used X-ray breakage of chromosomes to determine the distances between DNA markers as well as their order on the chromosome

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Methods for genome mapping

Happy mapping

- RH mapping have a few problems that arise from the *in vivo* step. Happy mapping overcomes these limitations, being an entirely *in vitro* technique

Methods for genome mapping

Map integration

- It is essential that the different mapping methods are integrated as each has its own strengths and weaknesses

Mapping and sequencing genomes

Sequencing genome

- The process of determining the exact order of nucleotides within a DNA or RNA molecule is called sequencing
- In 1970s, DNA sequencing was a very tedious process which involved determining only a few hundred nucleotides at a time

Mapping and sequencing genomes

High-throughput sequencing

- Recent advances in DNA sequencing have make it possible to sequence data very rapidly and at a substantially lower cost
- It mainly describe a number of different modern sequencing technologies to sequence genome

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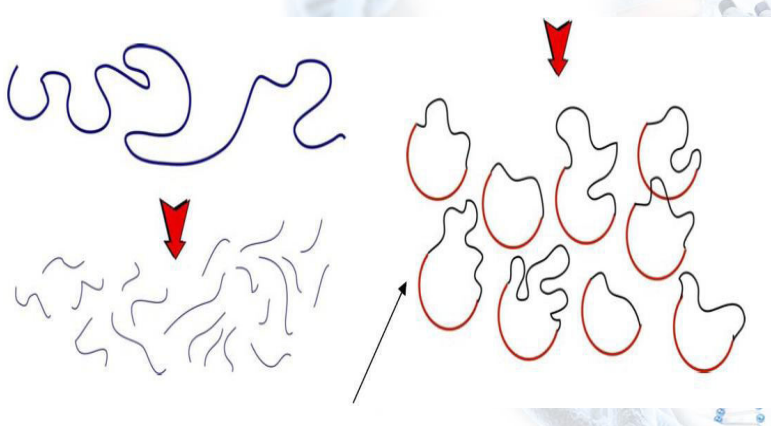
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Mapping and sequencing genomes

Shotgun sequencing

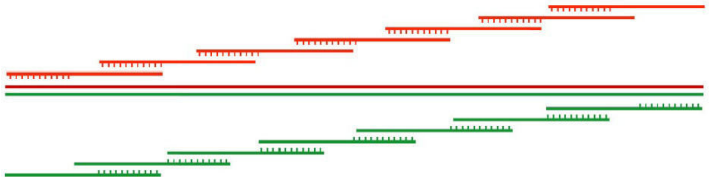
- Shotgun sequencing is used for sequencing long DNA strands
- In this strategy, DNA shredded into smaller fragments that can be sequenced individually and then reassembled into original order

Mapping and sequencing genomes



Shotgun library

Mapping and sequencing genomes



Shotgun library

Mapping and sequencing genomes

Clone-by-clone sequencing

- In clone-by-clone sequencing, a map of each chromosome of the genome is made before the DNA is split up into fragments for sequencing

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Comparative genomics

Orthologs and paralogs

- Orthologs are homologous genes in different organisms that encode proteins with the same function and which have evolved by direct vertical descent



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Applications of siRNA

- siRNA) is the method of choice to target specific genes for silencing and has provided immense potential as therapeutic tools

Comparative genomics

Comparative genomics of bacteria

- Genome of bacteria may be of variable in size. For example, it may vary from 0.49 Mb (*Nanoarchaeum equitans*) to 9.1 Mb (*Bradyrhizobium japonicum* or *Streptomyces*)

Comparative genomics

Comparative genomics of organelles

- Mitochondrial genomes exhibit an amazing structural diversity
- Gene transfer has occurred between mtDNA and nuclear DNA

Comparative genomics

Comparative genomics of eukaryotes

- The minimal eukaryotic genome is smaller than many bacterial genomes
- Comparative genomics can be used to identify genes and regulatory elements
- It gives insight into the evolution of key proteins

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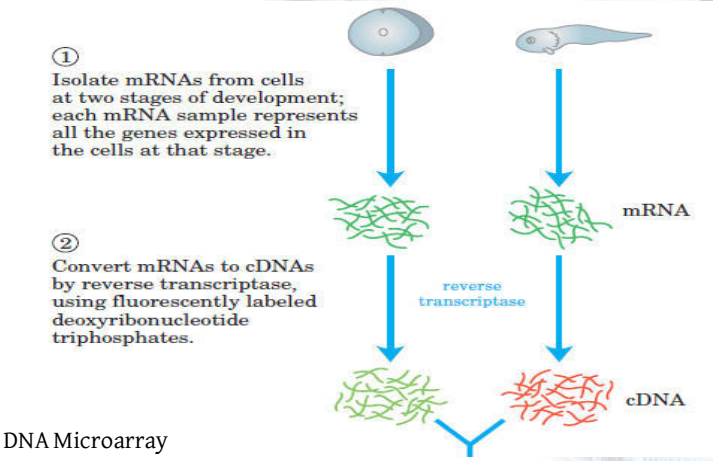
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Functional genomics and proteomics

DNA Microarrays

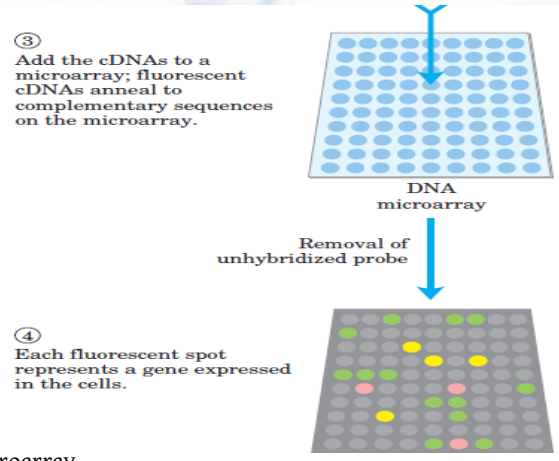
- Microarrays are used to study the expression of many genes at once. It involves the immobilization of thousands of gene sequences on glass slide that are then detected with complementary base pairing between the sample and the gene sequences on the chip

Functional genomics and proteomics



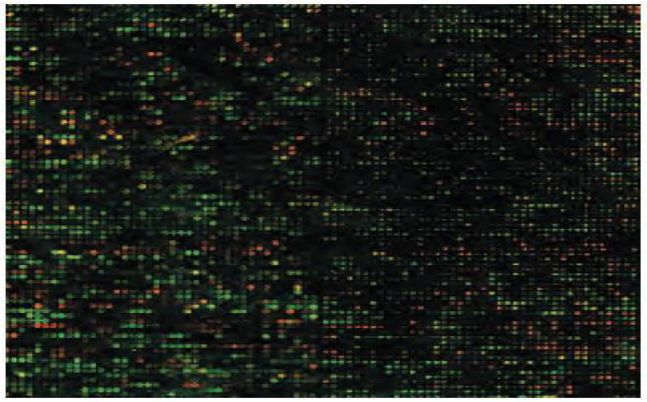
DNA Microarray

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DNA Microarray

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Enlarged image of a DNA Microarray

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Functional genomics and proteomics

DNA Microarrays

- Two major types of DNA array are used in expression analysis

i). Spotted DNA arrays

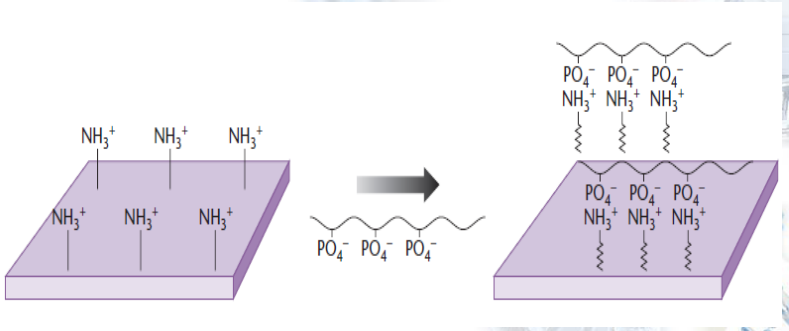
ii). Printed oligonucleotide chips

Functional genomics and proteomics

Spotted DNA arrays

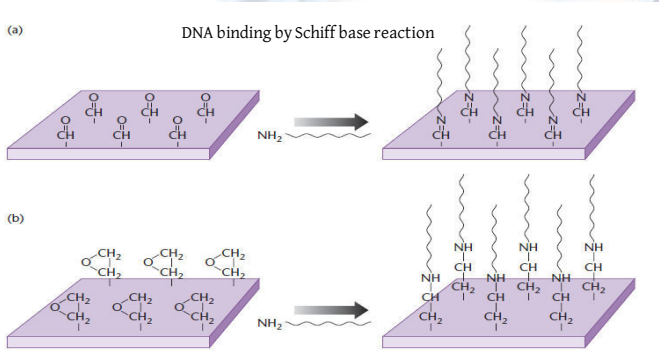
- Spotted DNA arrays are produced by printing DNA samples on treated microscope slides. It is made by transferring or spotting DNA clones or PCR product individually onto a solid support where they are immobilized

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DNA binding by electrostatic interactions

Functional genomics and proteomics



b)- Binding of DNA molecules to epoxy-derivatized surfaces

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Spotted DNA arrays

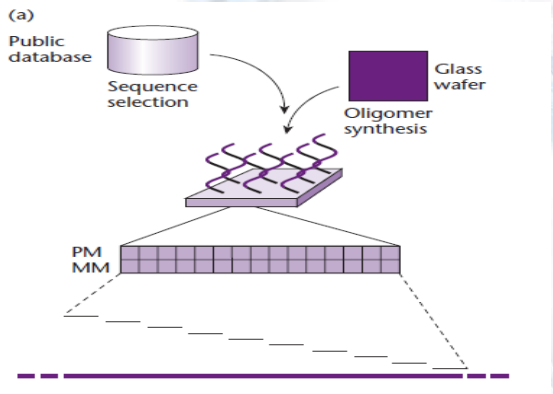
- Spotted DNA arrays are produced by printing DNA samples on treated microscope slides. It is made by transferring or spotting DNA clones or PCR product individually onto a solid support where they are immobilized

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Oligonucleotide chips

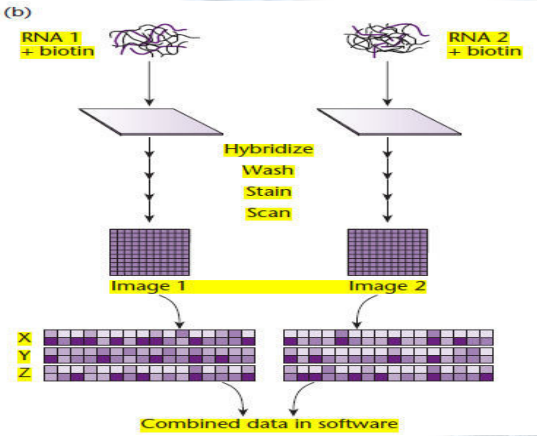
- Oligonucleotide chips are manufactured by in situ oligonucleotide synthesis

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Principle of chip manufacture

Functional genomics and proteomics



Principle of chip hybridization

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Applications of microarrays

Microbial gene expression analysis

- The predominate application of DNA microarray has been to measure gene expression levels

Applications of microarrays

Profiling in human disease

- Arrays have been used to investigate transcriptional profiles associated with human disease and to identify novel disease markers and potential new drug targets

Applications of microarrays

Genotyping

- Microarrays have been widely used as single nucleotide polymorphism (SNP) genotyping platforms

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Phage display

- Phage display is the technology that allows expression of exogenous polypeptides on the surface of phage particles

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Screening phage display libraries

- The most common screening method is based on enriching the phage clones with binding affinity for the target by a process called biopanning

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Applications of phage display

- Applications of phage display includes determination of protein to protein interactions, to determine enzyme specificity and to generate target specific antibodies

Functional genomics and proteomics

Knock outs

- A gene knock out is a genetic technique in which one of an organism's gene is made inoperative that is simply called as knocked out

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Knock ins

- In molecular cloning, a knock-in refers to a gene manipulation method that involves the insertion of a protein coding cDNA sequence at a particular locus in an organism's chromosome

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siRNA technology

- Small interfering RNA (siRNA) is the most commonly used RNA interference (RNAi) tool for inducing short-term silencing of protein coding genes. siRNA is a synthetic RNA duplex designed to specifically target a particular mRNA for degradation

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